

NOM: RASOLOFOMALALA

PRENOM: Maminiaina Angelo

NUMERO: 296I25

PROJET SQL – BD_AIR_MAROC

I. Création et préparation de la base

1. Base de données

```
C:\Windows\system32\cmd.exe - mysql -u root -p
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ANGELO>cd C:\Program Files\MariaDB 12.2\bin
C:\Program Files\MariaDB 12.2\bin>
C:\Program Files\MariaDB 12.2\bin>mysql -u root -p
Enter password: *****
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 4
Server version: 12.2.2-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| angelo   |
| cybercafe |
| information_schema |
| mysql    |
| performance_schema |
| sql_projet |
| sys      |
| test     |
+-----+
8 rows in set (0.001 sec)

MariaDB [(none)]> use sql_projet;
Database changed
MariaDB [sql_projet]> show tables;
Empty set (0.001 sec)

MariaDB [sql_projet]> CREATE TABLE IF NOT EXISTS PILOTE ( NUMPIL INT PRIMARY KEY,  NOMPIL VARCHAR(50),  VILLE VARCHAR(50),  SALAIRE DECIMAL(10,2));
Query OK, 0 rows affected (0.020 sec)

MariaDB [sql_projet]> CREATE TABLE IF NOT EXISTS AVION ( NUMAV INT PRIMARY KEY,  NOMAV VARCHAR(50),  CAPACITE INT,  VILLE VARCHAR(50));
Query OK, 0 rows affected (0.007 sec)

MariaDB [sql_projet]> CREATE TABLE IF NOT EXISTS VOL ( NUMVOL INT PRIMARY KEY,  NUMPIL INT,  NUMAV INT,  VILLE_DEP VARCHAR(50),  VILLE_ARR VARCHAR(50),  H_DEP TIME,  H_ARR TIME,  FOREIGN KEY (NUMPIL) REFERENCES PILOTE(NUMPIL),  FOREIGN KEY (NUMAV) REFERENCES AVION(NUMAV));
Query OK, 0 rows affected (0.010 sec)

MariaDB [sql_projet]> _
```

(Description: SHOW DATABASES + USE)

2. Tables

```
C:\Windows\system32\cmd.exe - mysql -u root -p
MariaDB [sql_projet]> CREATE TABLE IF NOT EXISTS PILOTE ( NUMPIL INT PRIMARY KEY, NOMPIL VARCHAR(50), VILLE VARCHAR(50), SALAIRE DECIMAL(10,2));
Query OK, 0 rows affected (0.020 sec)

MariaDB [sql_projet]> CREATE TABLE IF NOT EXISTS AVION ( NUMAV INT PRIMARY KEY, NOMAV VARCHAR(50), CAPACITE INT, VILLE VARCHAR(50));
Query OK, 0 rows affected (0.007 sec)

MariaDB [sql_projet]> CREATE TABLE IF NOT EXISTS VOL ( NUMVOL INT PRIMARY KEY, NUMPIL INT, NUMAV INT, VILLE_DEP VARCHAR(50), VILLE_ARR VARCHAR(50), H_DEP TIME, H_ARR TIME, FOREIGN KEY (NUMPIL) REFERENCES PILOTE(NUMPIL), FOREIGN KEY (NUMAV) REFERENCES AVION(NUMAV));
Query OK, 0 rows affected (0.010 sec)

MariaDB [sql_projet]> show tables;
+-----+
| Tables_in_sql_projet |
+-----+
| avion                 |
| pilote                |
| vol                   |
+-----+
3 rows in set (0.001 sec)

MariaDB [sql_projet]> _
```

(Description: CREATE TABLES + SHOW TABLES)

3. Table PILOTE

```
C:\Windows\system32\cmd.exe - mysql -u root -p
MariaDB [sql_projet]> CREATE TABLE IF NOT EXISTS AVION ( NUMAV INT PRIMARY KEY, NOMAV VARCHAR(50), CAPACITE INT, VILLE VARCHAR(50));
Query OK, 0 rows affected (0.007 sec)

MariaDB [sql_projet]> CREATE TABLE IF NOT EXISTS VOL ( NUMVOL INT PRIMARY KEY, NUMPIL INT, NUMAV INT, VILLE_DEP VARCHAR(50), VILLE_ARR VARCHAR(50), H_DEP TIME, H_ARR TIME, FOREIGN KEY (NUMPIL) REFERENCES PILOTE(NUMPIL), FOREIGN KEY (NUMAV) REFERENCES AVION(NUMAV));
Query OK, 0 rows affected (0.010 sec)

MariaDB [sql_projet]> show tables;
+-----+
| Tables_in_sql_projet |
+-----+
| avion                 |
| pilote                |
| vol                   |
+-----+
3 rows in set (0.001 sec)

MariaDB [sql_projet]> INSERT INTO PILOTE (NUMPIL, NOMPIL, VILLE, SALAIRE) VALUES
-> (1, 'Alami Amine', 'Casablanca', 25000),
-> (2, 'Bennani Youssef', 'Meknes', 22000),
-> (3, 'Chraïbi Omar', 'Marrakech', 18000),
-> (4, 'Daoudi Mehdi', 'Rabat', 21000),
-> (5, 'El Fassi Karim', 'Fes', 19500),
-> (6, 'Filali Hamza', 'Meknes', 23000),
-> (7, 'Ghazali Anas', 'Agadir', 17000),
-> (8, 'Habibi Said', 'Tanger', 20500),
-> (9, 'Idrissi Khalid', 'Oujda', 19000),
-> (10, 'Jebli Hassan', 'Guelmim', 16000),
-> (11, 'Kabbaj Ali', 'Casablanca', 27000),
-> (12, 'Lahlou Driss', 'Meknes', 24000),
-> (13, 'Mansouri Walid', 'Marrakech', 21500),
-> (14, 'Naji Brahim', 'Rabat', 18500),
-> (15, 'Ouazzani Adil', 'Fes', 20000),
-> (16, 'Pacha Reda', 'Agadir', 17500),
-> (17, 'Qasmi Mourad', 'Tanger', 22500),
-> (18, 'Rami Zakaria', 'Oujda', 19800),
-> (19, 'Slaoui Fouad', 'Guelmim', 16500),
-> (20, 'Tazi Ismail', 'Casablanca', 26000),
-> (21, 'Ummadi Yassine', 'Marrakech', 20200),
-> (22, 'Vadi Younes', 'Meknes', 18800),
-> (23, 'Wahbi Othman', 'Rabat', 23500),
-> (24, 'Xenifi Samy', 'Fes', 19200),
-> (25, 'Yacoubi Nabil', 'Agadir', 17800),
-> (26, 'Ziani Jalal', 'Tanger', 21200),
-> (27, 'Ait Ali Kamal', 'Oujda', 18200),
-> (28, 'Berrada Omar', 'Guelmim', 15800),
-> (29, 'Chahid Amine', 'Casablanca', 24500),
-> (30, 'Didi Hassan', 'Marrakech', 19700);
Query OK, 30 rows affected (0.065 sec)
Records: 30 Duplicates: 0 Warnings: 0

MariaDB [sql_projet]> _
```

(Description: INSERT + SELECT PILOTE)

4. Table AVION

```
C:\Windows\system32\cmd.exe - mysql -u root -p
-> (16, 'Pacha Reda', 'Agadir', 17500),
-> (17, 'Qasmi Mourad', 'Tanger', 22500),
-> (18, 'Rami Zakaria', 'Oujda', 19800),
-> (19, 'Slaoui Fouad', 'Guelmim', 16500),
-> (20, 'Tazi Ismail', 'Casablanca', 26000),
-> (21, 'Ummadi Yassine', 'Marrakech', 20200),
-> (22, 'Vadi Younes', 'Meknes', 18800),
-> (23, 'Wahbi Othman', 'Rabat', 23500),
-> (24, 'Xenifi Samy', 'Fes', 19200),
-> (25, 'Yacoubi Nabil', 'Agadir', 17800),
-> (26, 'Ziani Jalal', 'Tanger', 21200),
-> (27, 'Ait Ali Kamal', 'Oujda', 18200),
-> (28, 'Berrada Omar', 'Guelmim', 15800),
-> (29, 'Chahid Amine', 'Casablanca', 24500),
-> (30, 'Didi Hassan', 'Marrakech', 19700);
Query OK, 30 rows affected (0.065 sec)
Records: 30 Duplicates: 0 Warnings: 0

MariaDB [sql_projet]> INSERT INTO AVION (NUMAV, NOMAV, CAPACITE, VILLE) VALUES
-> (1, 'Boeing 737', 180, 'Casablanca'),
-> (2, 'Airbus A320', 160, 'Marrakech'),
-> (3, 'Boeing 777', 396, 'Casablanca'),
-> (4, 'Airbus A321', 228, 'Marrakech'),
-> (5, 'ATR 72', 70, 'Fes'),
-> (6, 'Boeing 787', 330, 'Rabat'),
-> (7, 'Airbus A350', 410, 'Agadir'),
-> (8, 'Boeing 737', 180, 'Tanger'),
-> (9, 'Airbus A320', 150, 'Oujda'),
-> (10, 'ATR 72', 70, 'Guelmim'),
-> (11, 'Airbus A330', 300, 'Meknes'),
-> (12, 'Boeing 767', 250, 'Guelmim'),
-> (13, 'Airbus A319', 140, 'Casablanca'),
-> (14, 'Boeing 757', 200, 'Marrakech'),
-> (15, 'CRJ 900', 90, 'Rabat'),
-> (16, 'Airbus A340', 380, 'Fes'),
-> (17, 'Boeing 727', 150, 'Agadir'),
-> (18, 'Airbus A310', 210, 'Tanger'),
-> (19, 'Boeing 707', 140, 'Meknes'),
-> (20, 'ATR 42', 45, 'Oujda'),
-> (21, 'Airbus A220', 130, 'Guelmim'),
-> (22, 'Boeing 717', 110, 'Casablanca'),
-> (23, 'Airbus A300', 260, 'Marrakech'),
-> (24, 'Boeing 747-8', 467, 'Casablanca'),
-> (25, 'Airbus A380-800', 525, 'Marrakech'),
-> (26, 'Boeing 737 Max', 170, 'Rabat'),
-> (27, 'Falcon 900', 14, 'Fes'),
-> (28, 'Gulfstream G650', 19, 'Agadir'),
-> (29, 'Airbus A320neo', 186, 'Tanger'),
-> (30, 'Boeing 737-800', 189, 'Meknes'),
-> (101, 'Reference', 300, 'Casablanca');
```

(Description: INSERT + SELECT AVION)

5. Table VOL

```
C:\Windows\system32\cmd.exe - mysql -u root -p
-> (28, 'Gulfstream G650', 19, 'Agadir'),
-> (29, 'Airbus A320neo', 186, 'Tanger'),
-> (30, 'Boeing 737-800', 189, 'Meknes'),
-> (101, 'Reference', 300, 'Casablanca');
Query OK, 31 rows affected (0.002 sec)
Records: 31 Duplicates: 0 Warnings: 0

MariaDB [sql_projet]> INSERT INTO VOL (NUMVOL, NUMPIL, NUMAV, VILLE_DEP, VILLE_ARR, H_DEP, H_ARR) VALUES
-> (101, 1, 1, 'Casablanca', 'Rabat', '08:00:00', '08:45:00'),
-> (102, 2, 2, 'Marrakech', 'Casablanca', '09:00:00', '10:00:00'),
-> (103, 3, 3, 'Casablanca', 'Tanger', '10:30:00', '11:30:00'),
-> (104, 4, 4, 'Marrakech', 'Agadir', '12:00:00', '12:45:00'),
-> (105, 5, 5, 'Fes', 'Oujda', '14:00:00', '14:45:00'),
-> (106, 6, 6, 'Rabat', 'Meknes', '15:30:00', '16:00:00'),
-> (107, 7, 7, 'Agadir', 'Casablanca', '17:00:00', '18:15:00'),
-> (108, 8, 8, 'Tanger', 'Fes', '19:00:00', '19:45:00'),
-> (109, 9, 9, 'Oujda', 'Casablanca', '07:30:00', '09:00:00'),
-> (110, 10, 10, 'Guelmim', 'Agadir', '08:15:00', '09:00:00'),
-> (111, 11, 11, 'Meknes', 'Rabat', '10:00:00', '10:30:00'),
-> (112, 12, 12, 'Guelmim', 'Casablanca', '11:15:00', '13:00:00'),
-> (113, 13, 13, 'Casablanca', 'Marrakech', '13:30:00', '14:30:00'),
-> (114, 14, 14, 'Marrakech', 'Oujda', '15:00:00', '16:30:00'),
-> (115, 15, 15, 'Rabat', 'Fes', '16:45:00', '17:15:00'),
-> (116, 16, 16, 'Fes', 'Casablanca', '18:00:00', '19:00:00'),
-> (117, 17, 17, 'Agadir', 'Tanger', '19:30:00', '21:00:00'),
-> (118, 18, 18, 'Tanger', 'Marrakech', '21:15:00', '22:30:00'),
-> (119, 19, 19, 'Meknes', 'Guelmim', '06:45:00', '08:15:00'),
-> (120, 20, 20, 'Oujda', 'Fes', '08:30:00', '09:15:00'),
-> (121, 21, 21, 'Guelmim', 'Casablanca', '10:45:00', '12:30:00'),
-> (122, 22, 22, 'Casablanca', 'Agadir', '13:00:00', '14:15:00'),
-> (123, 23, 23, 'Marrakech', 'Rabat', '14:45:00', '15:45:00'),
-> (124, 24, 24, 'Casablanca', 'Tanger', '16:15:00', '17:15:00'),
-> (125, 25, 25, 'Marrakech', 'Casablanca', '17:45:00', '18:45:00'),
-> (126, 1, 3, 'Casablanca', 'Marrakech', '19:15:00', '20:15:00'),
-> (127, 2, 4, 'Marrakech', 'Guelmim', '20:45:00', '21:45:00'),
-> (128, 3, 9, 'Casablanca', 'Oujda', '22:15:00', '23:45:00'),
-> (129, 4, 16, 'Fes', 'Marrakech', '07:00:00', '08:15:00'),
-> (130, 5, 24, 'Casablanca', 'Agadir', '09:30:00', '10:45:00');
```

(Description: INSERT + SELECT VOL)

II. Résolution des questions

Question 1

Donnez la liste des avions dont la capacité est supérieure à 350 passagers.

SQL:

```
SELECT * FROM AVION WHERE capacite > 350;
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
+----+-----+-----+-----+
| 27 | Falcon 900 | 14 | Fes |
| 28 | Gulfstream G650 | 19 | Agadir |
| 29 | Airbus A320neo | 186 | Tanger |
| 30 | Boeing 737-800 | 189 | Meknes |
| 101 | Reference | 300 | Casablanca |
+----+-----+-----+-----+
11 rows in set (0.069 sec)

MariaDB [sql_projet]> SELECT * FROM AVION WHERE Capacité > 350;
ERROR 1054 (42S22): Unknown column 'Capacité' in 'WHERE'
MariaDB [sql_projet]> SELECT * FROM AVION WHERE Capacite > 350;
+----+-----+-----+-----+
| NUMAV | NOMAV | CAPACITE | VILLE |
+----+-----+-----+-----+
| 3 | Boeing 777 | 396 | Casablanca |
| 7 | Airbus A350 | 410 | Agadir |
| 16 | Airbus A340 | 380 | Fes |
| 24 | Boeing 747-8 | 467 | Casablanca |
| 25 | Airbus A380-800 | 525 | Marrakech |
+----+-----+-----+-----+
5 rows in set (0.001 sec)

MariaDB [sql_projet]>
```

Question 2

Quels sont les numéros et noms des avions localisés à Marrakech ?

SQL:

```
SELECT numav, nomav FROM AVION WHERE localisation = 'Marrakech';
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
+----+-----+-----+-----+
| 27 | Falcon 900 | 14 | Fes |
| 28 | Gulfstream G650 | 19 | Agadir |
| 29 | Airbus A320neo | 186 | Tanger |
| 30 | Boeing 737-800 | 189 | Meknes |
| 101 | Reference | 300 | Casablanca |
+----+-----+-----+-----+
31 rows in set (0.069 sec)

MariaDB [sql_projet]> SELECT * FROM AVION WHERE Capacité > 350;
ERROR 1054 (42S22): Unknown column 'Capacité' in 'WHERE'
MariaDB [sql_projet]> SELECT * FROM AVION WHERE Capacite > 350;
+----+-----+-----+-----+
| NUMAV | NOMAV | CAPACITE | VILLE |
+----+-----+-----+-----+
| 3 | Boeing 777 | 396 | Casablanca |
| 7 | Airbus A350 | 410 | Agadir |
| 16 | Airbus A340 | 380 | Fes |
| 24 | Boeing 747-8 | 467 | Casablanca |
| 25 | Airbus A380-800 | 525 | Marrakech |
+----+-----+-----+-----+
5 rows in set (0.001 sec)

MariaDB [sql_projet]> SELECT NumAv, Nom FROM AVION WHERE Localisation = 'Marrakech';
ERROR 1054 (42S22): Unknown column 'Nom' in 'SELECT'
MariaDB [sql_projet]> SELECT NUMAV, NOMAV FROM AVION WHERE VILLE = 'Marrakech';
+----+-----+
| NUMAV | NOMAV |
+----+-----+
| 2 | Airbus A320 |
| 4 | Airbus A321 |
| 14 | Boeing 757 |
| 23 | Airbus A300 |
| 25 | Airbus A380-800 |
+----+-----+
5 rows in set (0.001 sec)

MariaDB [sql_projet]> _
```

Question 3

Quels sont les numéros des pilotes en service et les villes de départ de leurs vols ?

SQL:

```
SELECT DISTINCT numpil, ville_dep FROM VOL;
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
5 rows in set (0.001 sec)

MariaDB [sql_projet]> SELECT NumAv, Nom FROM AVION WHERE Localisation = 'Marrakech';
ERROR 1054 (42S22): Unknown column 'Nom' in 'SELECT'
MariaDB [sql_projet]> SELECT NUMAV, NOMAV FROM AVION WHERE VILLE = 'Marrakech';
+-----+-----+
| NUMAV | NOMAV |
+-----+-----+
| 2     | Airbus A320 |
| 4     | Airbus A321 |
| 14    | Boeing 757 |
| 23    | Airbus A300 |
| 25    | Airbus A380-800 |
+-----+-----+
5 rows in set (0.001 sec)

MariaDB [sql_projet]> SELECT DISTINCT NUMPIL, VILLE_DEP FROM VOL;
+-----+-----+
| NUMPIL | VILLE_DEP |
+-----+-----+
| 1     | Casablanca |
| 2     | Marrakech |
| 3     | Casablanca |
| 4     | Marrakech |
| 5     | Fes |
| 6     | Rabat |
| 7     | Agadir |
| 8     | Tanger |
| 9     | Oujda |
| 10    | Guelmim |
| 11    | Meknes |
| 12    | Guelmim |
| 13    | Casablanca |
| 14    | Marrakech |
| 15    | Rabat |
| 16    | Fes |
| 17    | Agadir |
| 18    | Tanger |
| 19    | Meknes |
| 20    | Oujda |
| 21    | Guelmim |
| 22    | Casablanca |
| 23    | Marrakech |
| 24    | Casablanca |
| 25    | Marrakech |
| 4     | Fes |
| 5     | Casablanca |
+-----+-----+
27 rows in set (0.003 sec)

MariaDB [sql_projet]>
```

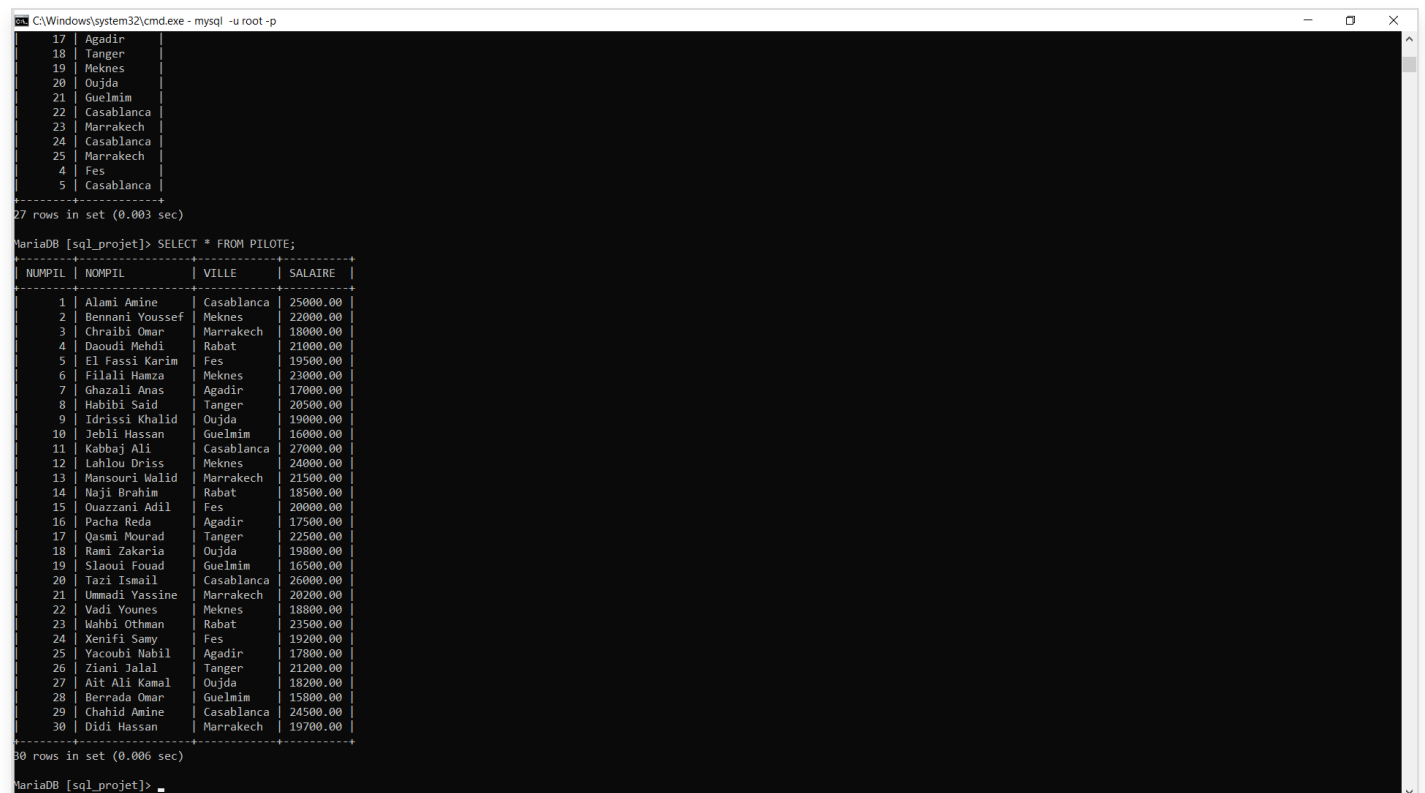
Question 4

Donnez toutes les informations sur les pilotes de la compagnie.

SQL:

```
SELECT * FROM PILOTE;
```

Résultat:



The screenshot shows a terminal window with a MySQL command prompt. The user has executed the query `SELECT * FROM PILOTE;` and the results are displayed in a table format. The table has four columns: `NUMPIL`, `NOWPIL`, `VILLE`, and `SALAIRE`. The results show 30 rows of pilot data, including names, IDs, cities, and salaries.

NUMPIL	NOWPIL	VILLE	SALAIRE
1	Alami Amine	Casablanca	25000.00
2	Bennani Youssef	Meknes	22000.00
3	Chraïbi Omar	Marrakech	18000.00
4	Daoudi Mehdi	Rabat	21000.00
5	El Fassi Karim	Fes	19500.00
6	Filali Hamza	Meknes	23000.00
7	Ghazali Anas	Agadir	17000.00
8	Habibi Said	Tanger	20500.00
9	Idrissi Khalid	Oujda	19000.00
10	Jebli Hassan	Guelmim	16000.00
11	Kabbaj Ali	Casablanca	27000.00
12	Lahlou Driss	Meknes	24000.00
13	Mansouri Walid	Marrakech	21500.00
14	Naji Brahim	Rabat	18500.00
15	Ouazzani Adil	Fes	20000.00
16	Pacha Reda	Agadir	17500.00
17	Qasmi Mourad	Tanger	22500.00
18	Rami Zakaria	Oujda	19800.00
19	Slaoui Fouad	Guelmim	16500.00
20	Tazi Ismail	Casablanca	26000.00
21	Ummadi Yassine	Marrakech	20200.00
22	Vadi Younes	Meknes	18800.00
23	Wahbi Othman	Rabat	23500.00
24	Xenifi Samy	Fes	19200.00
25	Yacoubi Nabil	Agadir	17800.00
26	Ziani Jalal	Tanger	21200.00
27	Ait Ali Kamal	Oujda	18200.00
28	Berrada Omar	Guelmim	15800.00
29	Chahid Amine	Casablanca	24500.00
30	Didi Hassan	Marrakech	19700.00

Question 5

Quel est le nom des pilotes domiciliés à Meknès dont le salaire est supérieur à 20000 DH ?

SQL:

```
SELECT nompil FROM PILOTE WHERE adresse = 'Meknès' AND salaire > 20000;
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
5 | Casablanca |
-----+-----
27 rows in set (0.003 sec)

MariaDB [sql_projet]> SELECT * FROM PILOTE;
+-----+-----+-----+-----+
| NUMPIL | NOMPIL | VILLE | SALAIRE |
+-----+-----+-----+-----+
| 1 | Alami Amine | Casablanca | 25000.00 |
| 2 | Bennani Youssef | Meknes | 22000.00 |
| 3 | Chraïbi Omar | Marrakech | 18000.00 |
| 4 | Daoudi Mehdi | Rabat | 21000.00 |
| 5 | El Fassi Karim | Fes | 19500.00 |
| 6 | Filali Hamza | Meknes | 23000.00 |
| 7 | Ghazali Anas | Agadir | 17000.00 |
| 8 | Habihi Said | Tanger | 20500.00 |
| 9 | Idrissi Khalid | Oujda | 19000.00 |
| 10 | Jebli Hassan | Guelmim | 16000.00 |
| 11 | Kabbaj Ali | Casablanca | 27000.00 |
| 12 | Lahlou Driss | Meknes | 24000.00 |
| 13 | Mansouri Walid | Marrakech | 21500.00 |
| 14 | Naji Brahim | Rabat | 18500.00 |
| 15 | Ouazzani Adil | Fes | 20000.00 |
| 16 | Pacha Reda | Agadir | 17500.00 |
| 17 | Qasmi Mourad | Tanger | 22500.00 |
| 18 | Rami Zakaria | Oujda | 19800.00 |
| 19 | Slaoui Fouad | Guelmim | 16500.00 |
| 20 | Tozi Ismail | Casablanca | 26000.00 |
| 21 | Ummadi Yassine | Marrakech | 20200.00 |
| 22 | Vadi Younes | Meknes | 18800.00 |
| 23 | Wahbi Othman | Rabat | 23500.00 |
| 24 | Xenifi Samy | Fes | 19200.00 |
| 25 | Yacoubi Nabil | Agadir | 17800.00 |
| 26 | Ziani Jalal | Tanger | 21200.00 |
| 27 | Ait Ali Kamal | Oujda | 18200.00 |
| 28 | Berrada Omar | Guelmim | 15800.00 |
| 29 | Chahid Amine | Casablanca | 24500.00 |
| 30 | Didi Hassan | Marrakech | 19700.00 |
+-----+-----+-----+-----+
30 rows in set (0.006 sec)

MariaDB [sql_projet]> SELECT NOMPIL FROM PILOTE WHERE VILLE = 'Meknès' AND SALAIRE > 20000;
+-----+
| NOMPIL |
+-----+
| Bennani Youssef |
| Filali Hamza |
| Lahlou Driss |
+-----+
3 rows in set (0.002 sec)

MariaDB [sql_projet]>
```

Question 6

Quels sont les avions (numéro et nom) localisés à Marrakech ou dont la capacité est inférieure à 350 passagers ?

SQL:

```
SELECT numav, nomav FROM AVION WHERE localisation = 'Marrakech' OR capacite < 350;
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
+----+-----+-----+-----+
| 27 | Ait Ali Kamal | Oujda | 18200.00 |
| 28 | Berrada Omar | Guelmim | 15800.00 |
| 29 | Chahid Amine | Casablanca | 24500.00 |
| 30 | Didi Hassan | Marrakech | 19700.00 |
+----+-----+-----+-----+
30 rows in set (0.006 sec)

MariaDB [sql_projet]> SELECT NOMPIL FROM PILOTE WHERE VILLE = 'Meknès' AND SALAIRE > 20000;
+-----+
| NOMPIL |
+-----+
| Bennani Youssef |
| Filali Hamza |
| Lahlou Driss |
+-----+
3 rows in set (0.002 sec)

MariaDB [sql_projet]> SELECT NUMAV, NOMAV FROM AVION WHERE VILLE = 'Marrakech' OR CAPACITE < 350;
+-----+-----+
| NUMAV | NOMAV |
+-----+-----+
| 1 | Boeing 737 |
| 2 | Airbus A320 |
| 4 | Airbus A321 |
| 5 | ATR 72 |
| 6 | Boeing 787 |
| 8 | Boeing 737 |
| 9 | Airbus A320 |
| 10 | ATR 72 |
| 11 | Airbus A330 |
| 12 | Boeing 767 |
| 13 | Airbus A319 |
| 14 | Boeing 757 |
| 15 | CRJ 900 |
| 17 | Boeing 727 |
| 18 | Airbus A310 |
| 19 | Boeing 707 |
| 20 | ATR 42 |
| 21 | Airbus A220 |
| 22 | Boeing 717 |
| 23 | Airbus A300 |
| 25 | Airbus A380-800 |
| 26 | Boeing 737 Max |
| 27 | Falcon 900 |
| 28 | Gulfstream G650 |
| 29 | Airbus A320neo |
| 30 | Boeing 737-800 |
| 101 | Reference |
+-----+-----+
27 rows in set (0.001 sec)

MariaDB [sql_projet]>
```

Question 7

Quels sont les numéros des pilotes qui ne sont pas en service ?

SQL:

```
SELECT numpil FROM PILOTE WHERE numpil NOT IN (SELECT numpil FROM VOL);
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
19 | Boeing 707 |
20 | ATR 42 |
21 | Airbus A220 |
22 | Boeing 717 |
23 | Airbus A300 |
25 | Airbus A300-800 |
26 | Boeing 737 Max |
27 | Falcon 900 |
28 | Gulfstream G650 |
29 | Airbus A320neo |
30 | Boeing 737-800 |
101 | Reference |
-----+-----
27 rows in set (0.001 sec)

MariaDB [sql_projet]> SELECT NUMPIL FROM PILOTE WHERE NUMPIL NOT IN (SELECT NUMPIL FROM VOL);
+-----+
| NUMPIL |
+-----+
| 26 |
| 27 |
| 28 |
| 29 |
| 30 |
+-----+
5 rows in set (0.005 sec)

MariaDB [sql_projet]>
```

Question 8

Donnez le numéro des vols effectués au départ de Marrakech par des pilotes de Meknès ?

SQL:

```
SELECT numvol FROM VOL V JOIN PILOTE P ON V.numpil = P.numpil WHERE  
V.ville_dep = 'Marrakech' AND P.adresse = 'Meknès';
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
19 | Boeing 707 |
20 | ATR 42 |
21 | Airbus A220 |
22 | Boeing 717 |
23 | Airbus A300 |
25 | Airbus A300-800 |
26 | Boeing 737 Max |
27 | Falcon 900 |
28 | Gulfstream G650 |
29 | Airbus A320neo |
30 | Boeing 737-800 |
101 | Reference |
-----+-----
27 rows in set (0.001 sec)

MariaDB [sql_projet]> SELECT NUMPIL FROM PILOTE WHERE NUMPIL NOT IN (SELECT NUMPIL FROM VOL);
+-----+
| NUMPIL |
+-----+
| 26 |
| 27 |
| 28 |
| 29 |
| 30 |
+-----+
5 rows in set (0.005 sec)

MariaDB [sql_projet]> Donnez le numéro des vols effectués au départ de Marrakech par des pilotes de Meknès
> ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'Donnez le numéro des vols effectués au départ de Marrakech par des pilotes...' at line 1
MariaDB [sql_projet]> SELECT NUMVOL FROM VOL JOIN PILOTE ON VOL.NUMPIL = PILOTE.NUMPIL WHERE VILLE_DEP = 'Marrakech' AND PILOTE.VILLE = 'Meknès';
+-----+
| NUMVOL |
+-----+
| 102 |
| 127 |
+-----+
2 rows in set (0.001 sec)

MariaDB [sql_projet]>
```

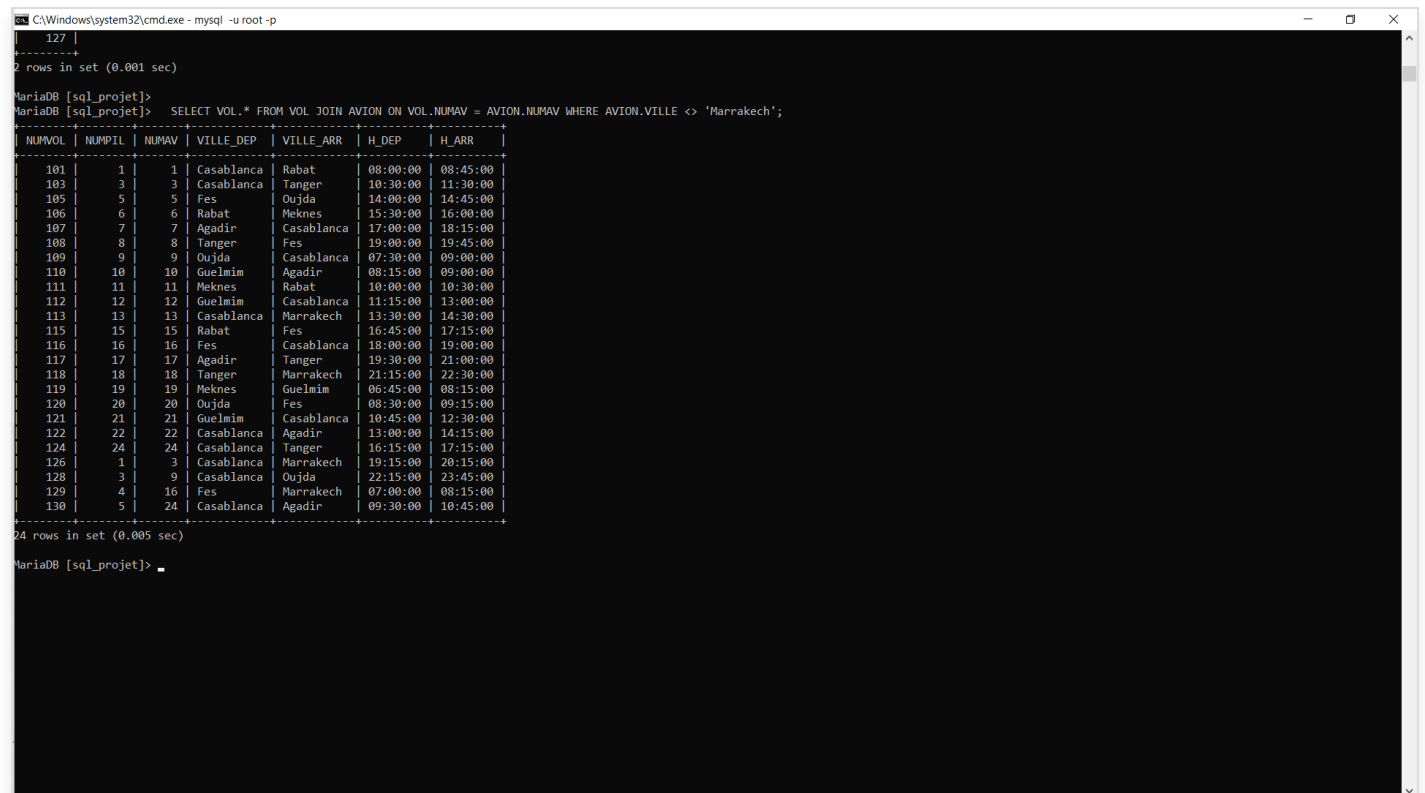
Question 9

Quels sont les vols effectués par un avion qui n'est pas localisé à Marrakech ?

SQL:

```
SELECT V.* FROM VOL V JOIN AVION A ON V.numav = A.numav WHERE A.localisation <> 'Marrakech';
```

Résultat:



The screenshot shows a Windows command prompt window with the title "C:\Windows\system32\cmd.exe - mysql -u root -p". The prompt shows the user has entered the password "127" and is now in the MySQL shell. The user has entered the query: `SELECT VOL.* FROM VOL JOIN AVION ON VOL.NUMAV = AVION.NUMAV WHERE AVION.VILLE <> 'Marrakech';`. The results are displayed in a table with 7 columns: NUMVOL, NUMPIL, NUMAV, VILLE_DEP, VILLE_ARR, H_DEP, and H_ARR. There are 24 rows of data, all representing flights from various cities to Marrakech. The prompt at the bottom shows the user has entered a period "." to exit the MySQL shell.

NUMVOL	NUMPIL	NUMAV	VILLE_DEP	VILLE_ARR	H_DEP	H_ARR
101	1	1	Casablanca	Rabat	08:00:00	08:45:00
103	3	3	Casablanca	Tanger	10:30:00	11:30:00
105	5	5	Fes	Oujda	14:00:00	14:45:00
106	6	6	Rabat	Meknes	15:30:00	16:00:00
107	7	7	Agadir	Casablanca	17:00:00	18:15:00
108	8	8	Tanger	Fes	19:00:00	19:45:00
109	9	9	Oujda	Casablanca	07:30:00	09:00:00
110	10	10	Guelmim	Agadir	08:15:00	09:00:00
111	11	11	Meknes	Rabat	10:00:00	10:30:00
112	12	12	Guelmim	Casablanca	11:15:00	13:00:00
113	13	13	Casablanca	Marrakech	13:30:00	14:30:00
115	15	15	Rabat	Fes	16:45:00	17:15:00
116	16	16	Fes	Casablanca	18:00:00	19:00:00
117	17	17	Agadir	Tanger	19:30:00	21:00:00
118	18	18	Tanger	Marrakech	21:15:00	22:30:00
119	19	19	Meknes	Guelmim	06:45:00	08:15:00
120	20	20	Oujda	Fes	08:30:00	09:15:00
121	21	21	Guelmim	Casablanca	10:45:00	12:30:00
122	22	22	Casablanca	Agadir	13:00:00	14:15:00
124	24	24	Casablanca	Tanger	16:15:00	17:15:00
126	1	3	Casablanca	Marrakech	19:15:00	20:15:00
128	3	9	Casablanca	Oujda	22:15:00	23:45:00
129	4	16	Fes	Marrakech	07:00:00	08:15:00
130	5	24	Casablanca	Agadir	09:30:00	10:45:00

Question 10

Quelles sont les villes desservies à partir de la ville d'arrivée d'un vol au départ de Guelmim ?

SQL:

```
SELECT DISTINCT ville_arr FROM VOL WHERE ville_dep IN (SELECT ville_arr FROM VOL WHERE ville_dep = 'Guelmim');
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
127 |
-----+-----
2 rows in set (0.001 sec)

MariaDB [sql_projet]>
MariaDB [sql_projet]> SELECT VOL.* FROM VOL JOIN AVION ON VOL.NUMAV = AVION.NUMAV WHERE AVION.VILLE <> 'Marrakech';
+-----+-----+-----+-----+-----+-----+-----+
| NUMVOL | NUMPIL | NUMAV | VILLE_DEP | VILLE_ARR | H_DEP | H_ARR |
+-----+-----+-----+-----+-----+-----+-----+
| 101    | 1      | 1     | Casablanca | Rabat     | 08:00:00 | 08:45:00 |
| 103    | 3      | 3     | Casablanca | Tanger    | 10:30:00 | 11:30:00 |
| 105    | 5      | 5     | Fes        | Oujda     | 14:00:00 | 14:45:00 |
| 106    | 6      | 6     | Rabat      | Meknes    | 15:30:00 | 16:00:00 |
| 107    | 7      | 7     | Agadir     | Casablanca | 17:00:00 | 18:15:00 |
| 108    | 8      | 8     | Tanger     | Fes       | 19:00:00 | 19:45:00 |
| 109    | 9      | 9     | Oujda      | Casablanca | 07:30:00 | 09:00:00 |
| 110    | 10     | 10    | Guelmim    | Agadir     | 08:15:00 | 09:00:00 |
| 111    | 11     | 11    | Meknes     | Rabat     | 10:00:00 | 10:30:00 |
| 112    | 12     | 12    | Guelmim    | Casablanca | 11:15:00 | 13:00:00 |
| 113    | 13     | 13    | Casablanca | Marrakech  | 13:30:00 | 14:30:00 |
| 115    | 15     | 15    | Rabat      | Fes       | 16:45:00 | 17:15:00 |
| 116    | 16     | 16    | Fes        | Casablanca | 18:00:00 | 19:00:00 |
| 117    | 17     | 17    | Agadir     | Tanger    | 19:30:00 | 21:00:00 |
| 118    | 18     | 18    | Tanger     | Marrakech  | 21:15:00 | 22:30:00 |
| 119    | 19     | 19    | Meknes     | Guelmim    | 06:45:00 | 08:15:00 |
| 120    | 20     | 20    | Oujda      | Fes       | 08:30:00 | 09:15:00 |
| 121    | 21     | 21    | Guelmim    | Casablanca | 10:45:00 | 12:30:00 |
| 122    | 22     | 22    | Casablanca | Agadir     | 13:00:00 | 14:15:00 |
| 124    | 24     | 24    | Casablanca | Tanger    | 16:15:00 | 17:15:00 |
| 126    | 1      | 3     | Casablanca | Marrakech  | 19:15:00 | 20:15:00 |
| 128    | 3      | 9     | Casablanca | Oujda     | 22:15:00 | 23:45:00 |
| 129    | 4      | 16    | Fes        | Marrakech  | 07:00:00 | 08:15:00 |
| 130    | 5      | 24    | Casablanca | Agadir     | 09:30:00 | 10:45:00 |
+-----+-----+-----+-----+-----+-----+-----+
24 rows in set (0.005 sec)

MariaDB [sql_projet]> SELECT DISTINCT VILLE_ARR FROM VOL WHERE VILLE_DEP IN (SELECT VILLE_ARR FROM VOL WHERE VILLE_DEP = 'Guelmim');
+-----+
| VILLE_ARR |
+-----+
| Rabat     |
| Tanger    |
| Casablanca |
| Marrakech  |
| Agadir     |
| Oujda     |
+-----+
6 rows in set (0.001 sec)

MariaDB [sql_projet]>
```

Question 11

Donner les noms des pilotes ainsi que les numéros des vols qu'ils assurent.

SQL:

```
SELECT P.nompil, V.numvol FROM PILOTE P JOIN VOL V ON P.numpil = V.numpil;
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
MariaDB [sql_projet]> SELECT DISTINCT VILLE_ARR FROM VOL WHERE VILLE_DEP IN (SELECT VILLE_ARR FROM VOL WHERE VILLE_DEP = 'Guelmim');
+-----+
| VILLE_ARR |
+-----+
| Rabat     |
| Tanger    |
| Casablanca |
| Marrakech  |
| Agadir     |
| Oujda      |
+-----+
6 rows in set (0.001 sec)

MariaDB [sql_projet]> SELECT NOMPIL, NUMVOL FROM PILOTE JOIN VOL ON PILOTE.NUMPIL = VOL.NUMPIL;
+-----+-----+
| NOMPIL | NUMVOL |
+-----+-----+
| Alami Amine | 101 |
| Alami Amine | 126 |
| Bennani Youssef | 102 |
| Bennani Youssef | 127 |
| Chraïbi Omar | 103 |
| Chraïbi Omar | 128 |
| Daoudi Mehdi | 104 |
| Daoudi Mehdi | 129 |
| El Fassi Karim | 105 |
| El Fassi Karim | 130 |
| Filali Hamza | 106 |
| Ghazali Anas | 107 |
| Habibi Said | 108 |
| Idrissi Khalid | 109 |
| Jebli Hassan | 110 |
| Kabbaj Ali | 111 |
| Lahlou Driss | 112 |
| Mansouri Walid | 113 |
| Naji Brahim | 114 |
| Ouazzani Adil | 115 |
| Pacha Reda | 116 |
| Qasbi Mounad | 117 |
| Rami Zakaria | 118 |
| Slaoui Fouad | 119 |
| Tazi Ismail | 120 |
| Ummadi Yassine | 121 |
| Vadi Younes | 122 |
| Wahbi Othman | 123 |
| Xenifi Samy | 124 |
| Yacoubi Nabil | 125 |
+-----+-----+
30 rows in set (0.001 sec)

MariaDB [sql_projet]>
```

Question 12

Donner les noms des pilotes et les noms des avions qu'ils pilotent.

SQL:

```
SELECT DISTINCT P.nompil, A.nomav FROM PILOTE P JOIN VOL V ON P.numpil = V.numpil JOIN AVION A ON V.numav = A.numav;
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
+-----+-----+
| Ouazzani Adil | 115 |
| Pacha Reda    | 116 |
| Qasmi Mourad  | 117 |
| Rami Zakaria  | 118 |
| Slaoui Fouad  | 119 |
| Tazi Ismail   | 120 |
| Ummadi Yassine | 121 |
| Vadi Younes   | 122 |
| Wahbi Othman  | 123 |
| Xenifi Samy   | 124 |
| Yacoubi Nabil | 125 |
+-----+-----+
30 rows in set (0.001 sec)

MariaDB [sql_projet]> SELECT NOMPIL, NOMAV FROM PILOTE JOIN VOL ON PILOTE.NUMPIL = VOL.NUMPIL JOIN AVION ON VOL.NUMAV = AVION.NUMAV;
+-----+-----+
| NOMPIL | NOMAV |
+-----+-----+
| Alami Amine | Boeing 737 |
| Bennani Youssef | Airbus A320 |
| Chraïbi Omar | Boeing 777 |
| Daoudi Mehdi | Airbus A321 |
| El Fassi Karim | ATR 72 |
| Filali Hamza | Boeing 787 |
| Ghazali Anas | Airbus A350 |
| Habibi Said | Boeing 737 |
| Idrissi Khalid | Airbus A320 |
| Jebli Hassan | ATR 72 |
| Kabbaj Ali | Airbus A330 |
| Lahlou Driss | Boeing 767 |
| Mansouri Walid | Airbus A319 |
| Naji Brahim | Boeing 757 |
| Ouazzani Adil | CRJ 900 |
| Pacha Reda | Airbus A340 |
| Qasmi Mourad | Boeing 727 |
| Rami Zakaria | Airbus A310 |
| Slaoui Fouad | Boeing 707 |
| Tazi Ismail | ATR 42 |
| Ummadi Yassine | Airbus A220 |
| Vadi Younes | Boeing 717 |
| Wahbi Othman | Airbus A300 |
| Xenifi Samy | Boeing 747-8 |
| Yacoubi Nabil | Airbus A380-800 |
| Alami Amine | Boeing 777 |
| Bennani Youssef | Airbus A321 |
| Chraïbi Omar | Airbus A320 |
| Daoudi Mehdi | Airbus A340 |
| El Fassi Karim | Boeing 747-8 |
+-----+-----+
30 rows in set (0.001 sec)

MariaDB [sql_projet]> _
```

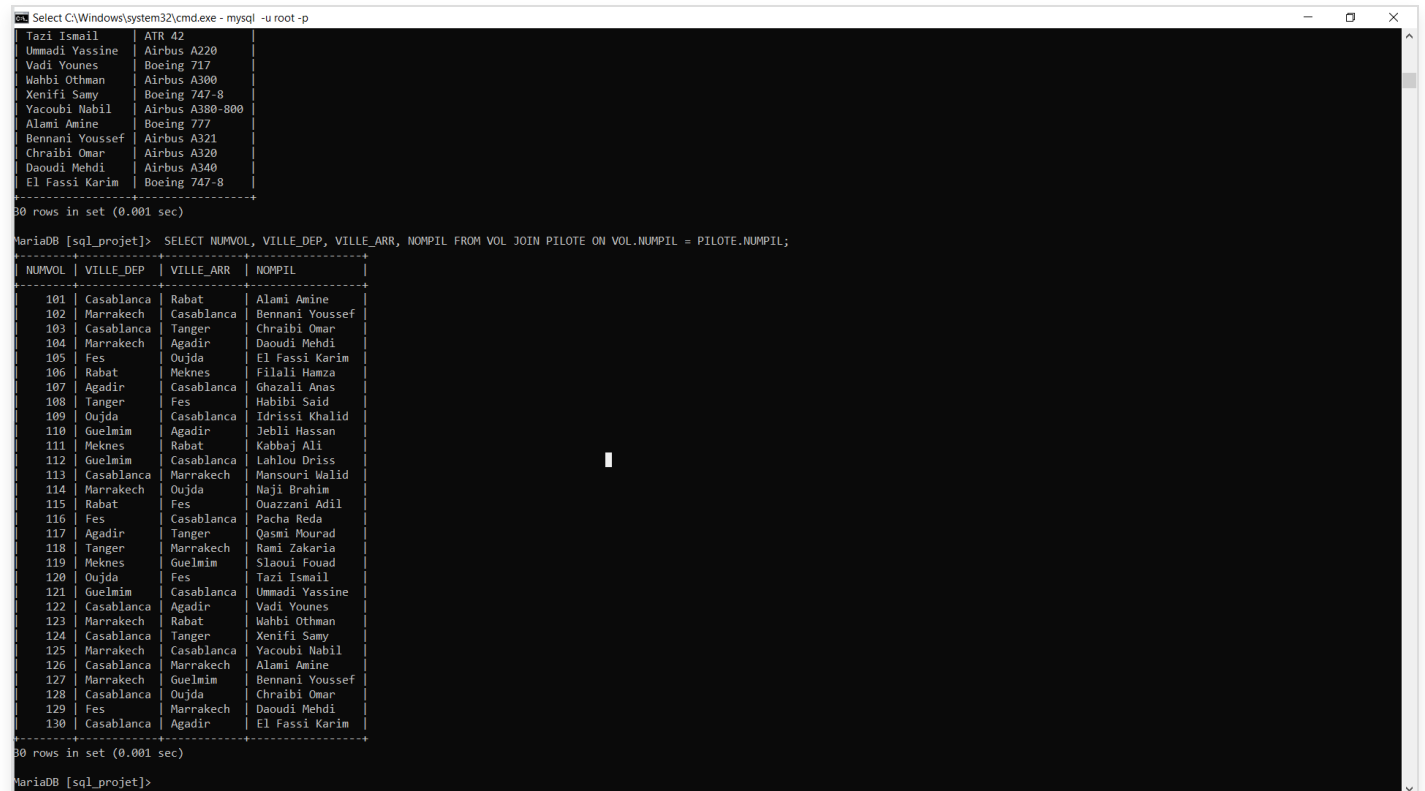

Question 13

Donner les vols (numéro, ville départ, ville arrivée) with le nom du pilote.

SQL:

```
SELECT V.numvol, V.ville_dep, V.ville_arr, P.nompil FROM VOL V JOIN PILOTE P
ON V.numpil = P.numpil;
```

Résultat:



```
Select C:\Windows\system32\cmd.exe - mysql -u root -p
MariaDB [sql_projet]> SELECT NUMVOL, VILLE_DEP, VILLE_ARR, NOMPIL FROM VOL JOIN PILOTE ON VOL.NUMPIL = PILOTE.NUMPIL;
```

NUMVOL	VILLE_DEP	VILLE_ARR	NOMPIL
101	Casablanca	Rabat	Alami Amine
102	Marrakech	Casablanca	Bennani Youssef
103	Casablanca	Tanger	Chraïbi Omar
104	Marrakech	Agadir	Daoudi Mehdi
105	Fes	Oujda	El Fassi Karim
106	Rabat	Meknes	Filali Hamza
107	Agadir	Casablanca	Ghazali Anas
108	Tanger	Fes	Habibi Said
109	Oujda	Casablanca	Idrissi Khalid
110	Guelmim	Agadir	Jebli Hassan
111	Meknes	Rabat	Kabbaj Ali
112	Guelmim	Casablanca	Lahlou Driss
113	Casablanca	Marrakech	Mansouri Walid
114	Marrakech	Oujda	Naji Brahim
115	Rabat	Fes	Ouazzani Adil
116	Fes	Casablanca	Pacha Reda
117	Agadir	Tanger	Qasmi Mourad
118	Tanger	Marrakech	Rami Zakaria
119	Meknes	Guelmim	Slaoui Fouad
120	Oujda	Fes	Tazi Ismail
121	Guelmim	Casablanca	Ummadi Yassine
122	Casablanca	Agadir	Vadi Younes
123	Marrakech	Rabat	Wahbi Othman
124	Casablanca	Tanger	Xenifi Samy
125	Marrakech	Casablanca	Yacoubi Nabil
126	Casablanca	Marrakech	Alami Amine
127	Marrakech	Guelmim	Bennani Youssef
128	Casablanca	Oujda	Chraïbi Omar
129	Fes	Marrakech	Daoudi Mehdi
130	Casablanca	Agadir	El Fassi Karim

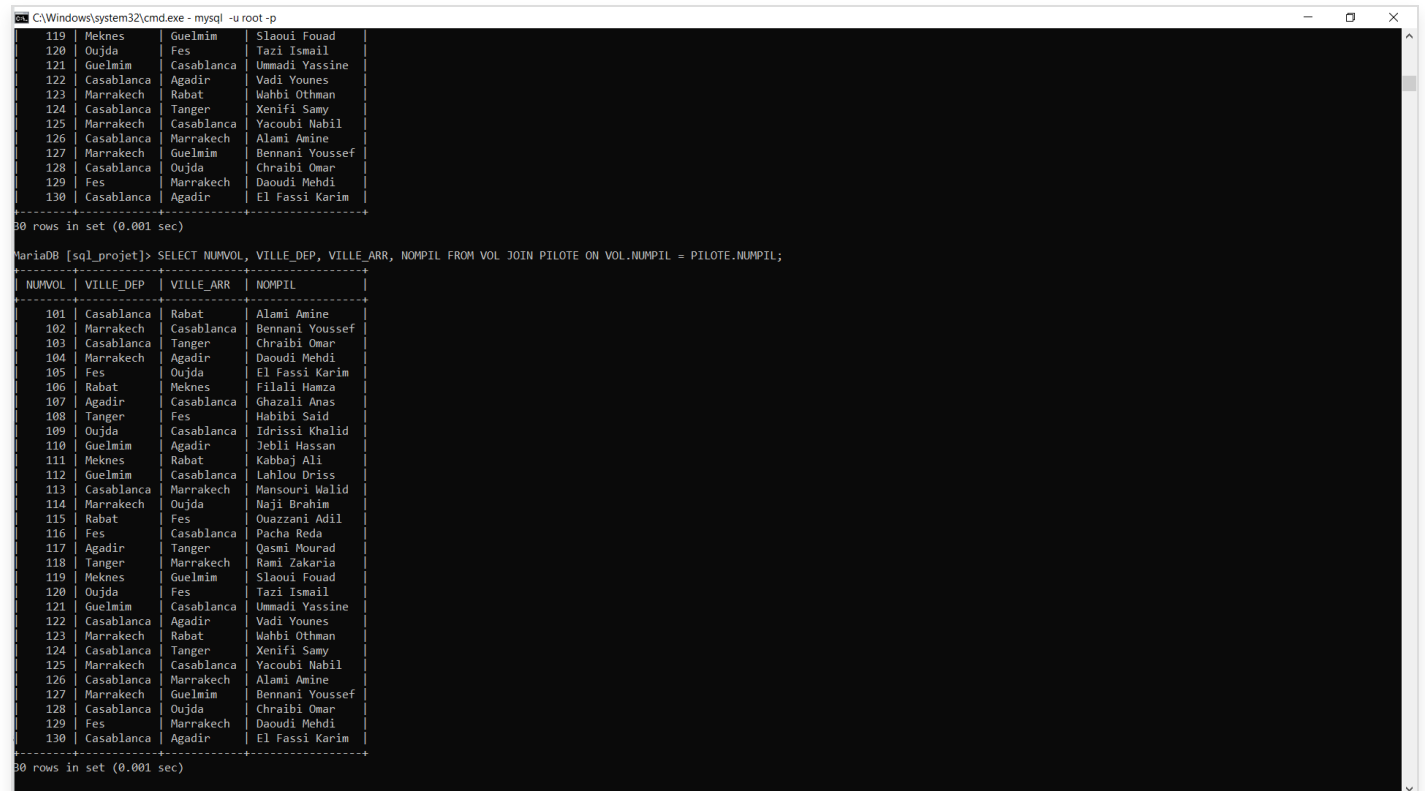
Question 14

Afficher les noms des avions utilisés pour les vols au départ de Casablanca.

SQL:

```
SELECT DISTINCT A.nomav FROM AVION A JOIN VOL V ON A.numav = V.numav WHERE  
V.ville_dep = 'Casablanca';
```

Résultat:



```
C:\Windows\system32\cmd.exe - mysql -u root -p  
119 | Meknes | Guelmim | Slaoui Fouad  
120 | Oujda | Fes | Tazi Ismail  
121 | Guelmim | Casablanca | Ummadi Yassine  
122 | Casablanca | Agadir | Vadi Younes  
123 | Marrakech | Rabat | Wahbi Othman  
124 | Casablanca | Tanger | Xenifi Samy  
125 | Marrakech | Casablanca | Yacoubi Nabil  
126 | Casablanca | Marrakech | Alami Amine  
127 | Marrakech | Guelmim | Bennani Youssef  
128 | Casablanca | Oujda | Chraïbi Omar  
129 | Fes | Marrakech | Daoudi Mehdi  
130 | Casablanca | Agadir | El Fassi Karim  
-----  
30 rows in set (0.001 sec)  
  
MariaDB [sql_projet]> SELECT NUMVOL, VILLE_DEP, VILLE_ARR, NOMPIL FROM VOL JOIN PILOTE ON VOL.NUMPIL = PILOTE.NUMPIL;  
+-----+-----+-----+-----+  
NUMVOL | VILLE_DEP | VILLE_ARR | NOMPIL  
+-----+-----+-----+-----+  
101 | Casablanca | Rabat | Alami Amine  
102 | Marrakech | Casablanca | Bennani Youssef  
103 | Casablanca | Tanger | Chraïbi Omar  
104 | Marrakech | Agadir | Daoudi Mehdi  
105 | Fes | Oujda | El Fassi Karim  
106 | Rabat | Meknes | Filali Hamza  
107 | Agadir | Casablanca | Ghazali Anas  
108 | Tanger | Fes | Habibi Said  
109 | Oujda | Casablanca | Idrissi Khalid  
110 | Guelmim | Agadir | Jebli Hassan  
111 | Meknes | Rabat | Kahbouj Ali  
112 | Guelmim | Casablanca | Lahlou Driss  
113 | Casablanca | Marrakech | Mansouri Walid  
114 | Marrakech | Oujda | Naji Brahim  
115 | Rabat | Fes | Ouazzani Adil  
116 | Fes | Casablanca | Pacha Reda  
117 | Agadir | Tanger | Qasmi Mourad  
118 | Tanger | Marrakech | Rami Zakaria  
119 | Meknes | Guelmim | Slaoui Fouad  
120 | Oujda | Fes | Tazi Ismail  
121 | Guelmim | Casablanca | Ummadi Yassine  
122 | Casablanca | Agadir | Vadi Younes  
123 | Marrakech | Rabat | Wahbi Othman  
124 | Casablanca | Tanger | Xenifi Samy  
125 | Marrakech | Casablanca | Yacoubi Nabil  
126 | Casablanca | Marrakech | Alami Amine  
127 | Marrakech | Guelmim | Bennani Youssef  
128 | Casablanca | Oujda | Chraïbi Omar  
129 | Fes | Marrakech | Daoudi Mehdi  
130 | Casablanca | Agadir | El Fassi Karim  
-----  
30 rows in set (0.001 sec)
```

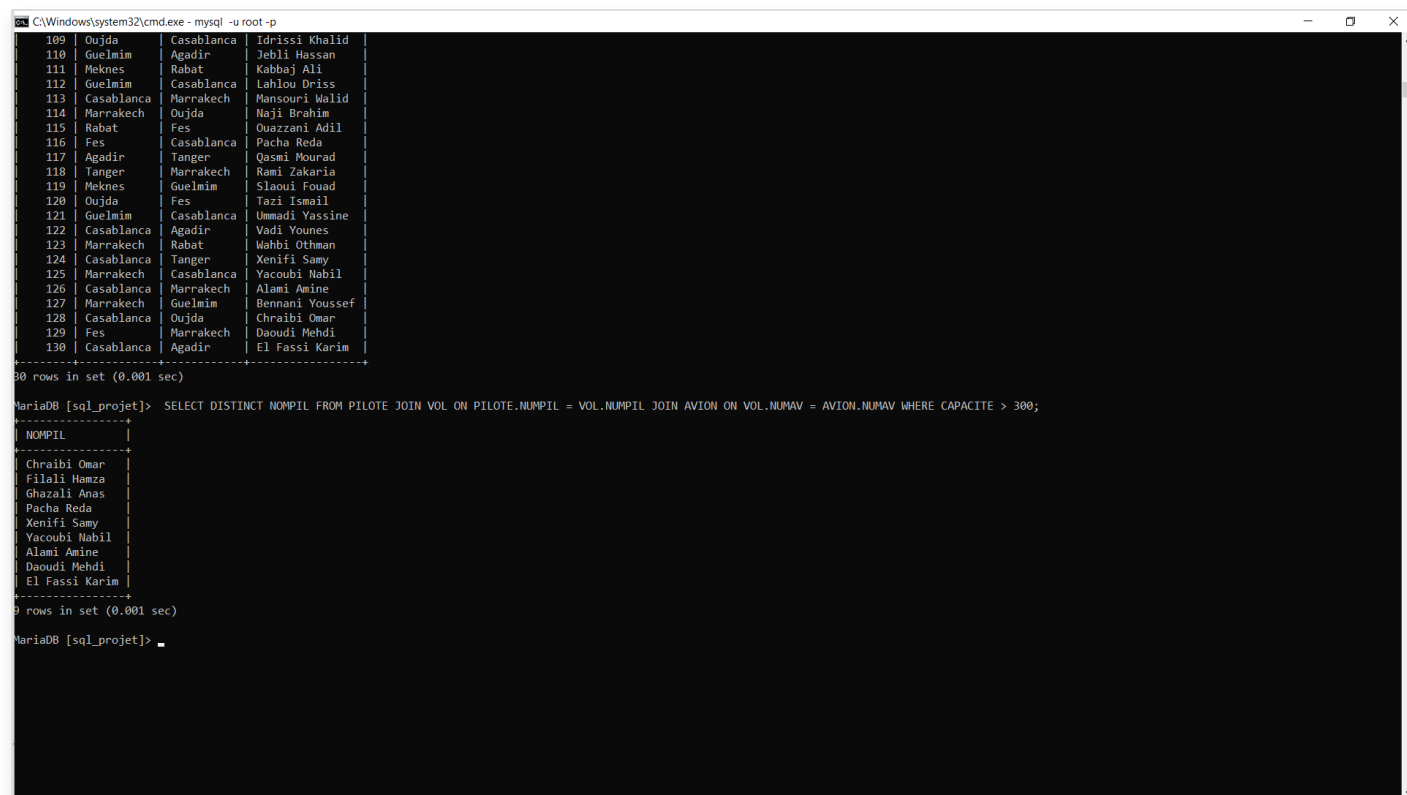
Question 15

Donner les noms des pilotes qui pilotent des avions de capacité supérieure à 300 passagers.

SQL:

```
SELECT DISTINCT P.nompil FROM PILOTE P JOIN VOL V ON P.numpil = V.numpil  
JOIN AVION A ON V.numav = A.numav WHERE A.capacite > 300;
```

Résultat:



The screenshot shows a MySQL command prompt window. The first query lists 30 rows of pilot data with columns for pilot ID, name, city, and aircraft type. The second query filters for pilots whose aircraft have a capacity greater than 300, returning 9 rows of pilot names.

```
C:\Windows\system32\cmd.exe - mysql -u root -p  
109 | Oujda | Casablanca | Idrissi Khalid  
110 | Guelmim | Agadir | Jebli Hassan  
111 | Meknes | Rabat | Kabbaj Ali  
112 | Guelmim | Casablanca | Lahlou Driss  
113 | Casablanca | Marrakech | Mansouri Walid  
114 | Marrakech | Oujda | Najji Brahim  
115 | Rabat | Fes | Ouezani Adil  
116 | Fes | Casablanca | Pacha Reda  
117 | Agadir | Tanger | Qasbi Mourad  
118 | Tanger | Marrakech | Rami Zakaria  
119 | Meknes | Guelmim | Slaoui Fouad  
120 | Oujda | Fes | Tazi Ismail  
121 | Guelmim | Casablanca | Ummadi Yassine  
122 | Casablanca | Agadir | Vadi Younes  
123 | Marrakech | Rabat | Wahbi Othman  
124 | Casablanca | Tanger | Xenifi Samy  
125 | Marrakech | Casablanca | Yacoubi Nabil  
126 | Casablanca | Marrakech | Alami Amine  
127 | Marrakech | Guelmim | Benmani Youssef  
128 | Casablanca | Oujda | Chraïbi Omar  
129 | Fes | Marrakech | Daoudi Mehdi  
130 | Casablanca | Agadir | El Fassi Karim  
30 rows in set (0.001 sec)  
  
MariaDB [sql_projet]> SELECT DISTINCT NOMPIL FROM PILOTE JOIN VOL ON PILOTE.NUMPIL = VOL.NUMPIL JOIN AVION ON VOL.NUMAV = AVION.NUMAV WHERE CAPACITE > 300;  
+-----+  
| NOMPIL |  
+-----+  
| Chraïbi Omar |  
| Filali Hamza |  
| Ghazali Anas |  
| Pacha Reda |  
| Xenifi Samy |  
| Yacoubi Nabil |  
| Alami Amine |  
| Daoudi Mehdi |  
| El Fassi Karim |  
+-----+  
9 rows in set (0.001 sec)  
  
MariaDB [sql_projet]> .
```

Question 16

Afficher les pilotes et les villes d'arrivée de leurs vols.

SQL:

```
SELECT P.nompil, V.ville_arr FROM PILOTE P JOIN VOL V ON P.numpil =  
V.numpil;
```

Résultat:

[Image non disponible]

Question 17

Donner les vols with les informations complètes des pilotes et des avions (jointure des 3 tables).

SQL:

```
SELECT * FROM VOL V JOIN PILOTE P ON V.numpil = P.numpil JOIN AVION A ON  
V.numav = A.numav;
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
+-----+
| NOMPIL |
+-----+
| Chraïbi Omar |
| Filali Hamza |
| Ghazali Anas |
| Pacha Reda |
| Xenifi Samy |
| Yacoubi Nabil |
| Alami Amine |
| Daoudi Mehdi |
| El Fassi Karim |
+-----+
9 rows in set (0.001 sec)

MariaDB [sql_projet]> SELECT NOMPIL, VILLE_ARR FROM PILOTE JOIN VOL ON PILOTE.NUMPIL = VOL.NUMPIL;
+-----+-----+
| NOMPIL | VILLE_ARR |
+-----+-----+
| Alami Amine | Rabat |
| Bennani Youssef | Casablanca |
| Chraïbi Omar | Tanger |
| Daoudi Mehdi | Agadir |
| El Fassi Karim | Oujda |
| Filali Hamza | Meknes |
| Ghazali Anas | Casablanca |
| Habibi Said | Fes |
| Idrissi Khalid | Casablanca |
| Jebli Hassan | Agadir |
| Kabbaj Ali | Rabat |
| Lahlou Driss | Casablanca |
| Mansouri Walid | Marrakech |
| Naji Brahim | Oujda |
| Ouazzani Adil | Fes |
| Pacha Reda | Casablanca |
| Qasmi Mourad | Tanger |
| Rami Zakaria | Marrakech |
| Slaoui Fouad | Guelmim |
| Tazi Ismail | Fes |
| Ummadi Yassine | Casablanca |
| Vadi Younes | Agadir |
| Wahbi Othman | Rabat |
| Xenifi Samy | Tanger |
| Yacoubi Nabil | Casablanca |
| Alami Amine | Marrakech |
| Bennani Youssef | Guelmim |
| Chraïbi Omar | Oujda |
| Daoudi Mehdi | Marrakech |
| El Fassi Karim | Agadir |
+-----+-----+
30 rows in set (0.000 sec)

MariaDB [sql_projet]> _
```

Question 18

Donner les noms des pilotes qui gagnent plus que le salaire moyen.

SQL:

```
SELECT nompil FROM PILOTE WHERE salaire > (SELECT AVG(salaire) FROM PILOTE);
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
+-----+
| Chraïbi Omar | Tanger |
| Daoudi Mehdi | Agadir |
| El Fassi Karim | Oujda |
| Filali Hamza | Meknes |
| Ghazali Anas | Casablanca |
| Habibi Said | Fes |
| Idrissi Khalid | Casablanca |
| Jebli Hassan | Agadir |
| Kabbaj Ali | Rabat |
| Lahlou Driss | Casablanca |
| Mansouri Walid | Marrakech |
| Naji Brahim | Oujda |
| Ouazzani Adil | Fes |
| Pacha Reda | Casablanca |
| Qasmi Mourad | Tanger |
| Rami Zakaria | Marrakech |
| Slaoui Fouad | Guelmim |
| Tazi Ismail | Fes |
| Ummadi Yassine | Casablanca |
| Vadi Younes | Agadir |
| Wahbi Othman | Rabat |
| Xenifi Samy | Tanger |
| Yacoubi Nabil | Casablanca |
| Alami Amine | Marrakech |
| Bennani Youssef | Guelmim |
| Chraïbi Omar | Oujda |
| Daoudi Mehdi | Marrakech |
| El Fassi Karim | Agadir |
+-----+
30 rows in set (0.000 sec)

MariaDB [sql_projet]> SELECT NOMPIL FROM PILOTE WHERE SALAIRE > (SELECT AVG(SALAIRE) FROM PILOTE);
+-----+
| NOMPIL |
+-----+
| Alami Amine |
| Bennani Youssef |
| Daoudi Mehdi |
| Filali Hamza |
| Habibi Said |
| Kabbaj Ali |
| Lahlou Driss |
| Mansouri Walid |
| Qasmi Mourad |
| Tazi Ismail |
| Wahbi Othman |
| Ziani Jalal |
| Chahid Amine |
+-----+
13 rows in set (0.002 sec)

MariaDB [sql_projet]>
```

Question 19

Donner les avions ayant une capacité supérieure à celle de l'avion numéro 101.

SQL:

```
SELECT * FROM AVION WHERE capacite > (SELECT capacite FROM AVION WHERE numav = 101);
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
+-----+
| Alami Amine |
| Bannani Youssef |
| Daoudi Mehdi |
| Filali Hamza |
| Habibi Said |
| Kabbej Ali |
| Lahlou Driss |
| Mansouri Walid |
| Qasmi Mourad |
| Tazi Ismail |
| Wahbi Othman |
| Ziani Jalal |
| Chahid Amine |
+-----+
13 rows in set (0.002 sec)

MariaDB [sql_projet]> SELECT * FROM AVION WHERE CAPACITE > (SELECT CAPACITE FROM AVION WHERE NUMAV = 101);
+-----+-----+-----+-----+
| NUMAV | NOMAV | CAPACITE | VILLE |
+-----+-----+-----+-----+
| 3 | Boeing 777 | 396 | Casablanca |
| 6 | Boeing 787 | 330 | Rabat |
| 7 | Airbus A350 | 410 | Agadir |
| 16 | Airbus A340 | 380 | Fes |
| 24 | Boeing 747-8 | 467 | Casablanca |
| 25 | Airbus A380-800 | 525 | Marrakech |
+-----+-----+-----+-----+
6 rows in set (0.001 sec)

MariaDB [sql_projet]>
```

Question 20

Donner les pilotes qui n'ont jamais effectué de vol.

SQL:

```
SELECT * FROM PILOTE WHERE numpil NOT IN (SELECT numpil FROM VOL);
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
+-----+
| Alami Amine |
| Bannani Youssef |
| Douadi Mehdi |
| Filali Hamza |
| Habibi Said |
| Kabbaj Ali |
| Lahlou Driss |
| Mansouri Walid |
| Qasmi Mourad |
| Tazi Ismail |
| Wahbi Othman |
| Ziani Jalal |
| Chahid Amine |
+-----+
13 rows in set (0.002 sec)

MariaDB [sql_projet]> SELECT * FROM AVION WHERE CAPACITE > (SELECT CAPACITE FROM AVION WHERE NUMAV = 101);
+-----+
| NUMAV | NOMAV | CAPACITE | VILLE |
+-----+
| 3 | Boeing 777 | 396 | Casablanca |
| 6 | Boeing 787 | 330 | Rabat |
| 7 | Airbus A350 | 410 | Agadir |
| 16 | Airbus A340 | 380 | Fes |
| 24 | Boeing 747-8 | 467 | Casablanca |
| 25 | Airbus A380-800 | 525 | Marrakech |
+-----+
6 rows in set (0.001 sec)

MariaDB [sql_projet]> SELECT NOMPIL FROM PILOTE WHERE NUMPIL NOT IN (SELECT NUMPIL FROM VOL);
+-----+
| NOMPIL |
+-----+
| Ziani Jalal |
| Ait Ali Kamal |
| Berrada Omar |
| Chahid Amine |
| Didi Hassen |
+-----+
5 rows in set (0.001 sec)

MariaDB [sql_projet]> _
```

Question 21

Donner les villes de départ des vols effectués par des pilotes habitant Marrakech.

SQL:

```
SELECT DISTINCT ville_dep FROM VOL V JOIN PILOTE P ON V.numpil = P.numpil  
WHERE P.adresse = 'Marrakech';
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p  
+-----+  
| Alami Amine |  
| Bennani Youssef |  
| Daoudi Mehdi |  
| Filali Hamza |  
| Habibi Said |  
| Kabbej Ali |  
| Lahlou Driss |  
| Mansouri Walid |  
| Qasmi Mourad |  
| Tazi Ismail |  
| Wahbi Othman |  
| Ziani Jalal |  
| Chahid Amine |  
+-----+  
13 rows in set (0.002 sec)  
  
MariaDB [sql_projet]> SELECT * FROM AVION WHERE CAPACITE > (SELECT CAPACITE FROM AVION WHERE NUMAV = 101);  
+-----+  
| NUMAV | NOMAV | CAPACITE | VILLE |  
+-----+  
| 3 | Boeing 777 | 396 | Casablanca |  
| 6 | Boeing 787 | 330 | Rabat |  
| 7 | Airbus A350 | 410 | Agadir |  
| 16 | Airbus A340 | 380 | Fes |  
| 24 | Boeing 747-8 | 467 | Casablanca |  
| 25 | Airbus A380-800 | 525 | Marrakech |  
+-----+  
6 rows in set (0.001 sec)  
  
MariaDB [sql_projet]> SELECT NOMPIL FROM PILOTE WHERE NUMPIL NOT IN (SELECT NUMPIL FROM VOL);  
+-----+  
| NOMPIL |  
+-----+  
| Ziani Jalal |  
| Ait Ali Kamal |  
| Berrada Omar |  
| Chahid Amine |  
| Didi Hassan |  
+-----+  
5 rows in set (0.001 sec)  
  
MariaDB [sql_projet]> SELECT DISTINCT VILLE_DEP FROM VOL JOIN PILOTE ON VOL.NUMPIL = PILOTE.NUMPIL WHERE PILOTE.VILLE = 'Marrakech';  
+-----+  
| VILLE_DEP |  
+-----+  
| Casablanca |  
| Guelmim |  
+-----+  
2 rows in set (0.001 sec)  
  
MariaDB [sql_projet]>
```


Question 22

Créer une vue contenant les vols with le nom du pilote et le nom de l'avion.

SQL:

```
CREATE VIEW vue_vols AS SELECT V.*, P.nompil, A.nomav FROM VOL V JOIN PILOTE P ON V.numpil = P.numpil JOIN AVION A ON V.numav = A.numav;
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
5 rows in set (0.001 sec)

MariaDB [sql_projet]> SELECT NOMPIL FROM PILOTE WHERE NOMPIL NOT IN (SELECT NUMPIL FROM VOL);
+-----+
| NOMPIL |
+-----+
| Ziani  |
| Ait Ali |
| Berrada |
| Chahid  |
| Didi    |
+-----+
5 rows in set (0.001 sec)

MariaDB [sql_projet]> SELECT DISTINCT VILLE_DEP FROM VOL JOIN PILOTE ON VOL.NUMPIL = PILOTE.NUMPIL WHERE PILOTE.VILLE = 'Marrakech';
+-----+
| VILLE_DEP |
+-----+
| Casablanca |
| Guelmim    |
+-----+
2 rows in set (0.001 sec)

MariaDB [sql_projet]> CREATE VIEW V_VOL_PIL_AV AS
-> SELECT V.NUMVOL, P.NOMPIL, A.NOMAV
-> FROM VOL V
-> JOIN PILOTE P ON V.NUMPIL = P.NUMPIL
-> JOIN AVION A ON V.NUMAV = A.NUMAV;
Query OK, 0 rows affected (0.006 sec)

MariaDB [sql_projet]>
```

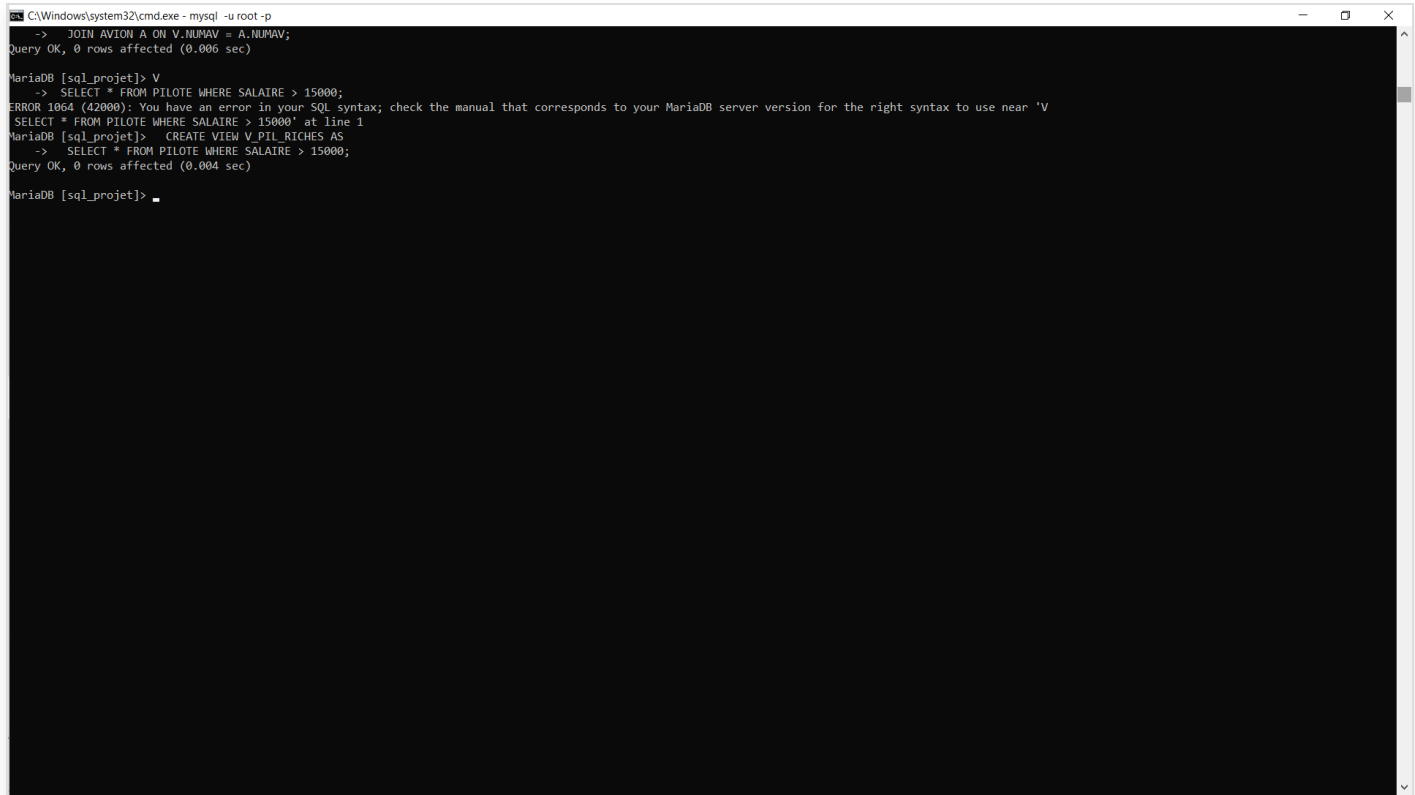
Question 23

Créer une vue des pilotes ayant un salaire supérieur à 15000 DH.

SQL:

```
CREATE VIEW vue_pilotes_riches AS SELECT * FROM PILOTE WHERE salaire > 15000;
```

Résultat:



```
C:\Windows\system32\cmd.exe - mysql -u root -p
-> JOIN AVION A ON V.NUMAV = A.NUMAV;
Query OK, 0 rows affected (0.006 sec)

MariaDB [sql_projet]> V
-> SELECT * FROM PILOTE WHERE SALAIRE > 15000;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'V
SELECT * FROM PILOTE WHERE SALAIRE > 15000' at line 1
MariaDB [sql_projet]> CREATE VIEW V.PIL_RICHES AS
-> SELECT * FROM PILOTE WHERE SALAIRE > 15000;
Query OK, 0 rows affected (0.004 sec)

MariaDB [sql_projet]> .
```

Question 24

Afficher les données de la vue créée précédemment.

SQL:

```
SELECT * FROM vue_pilotes_riches;
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
-> JOIN AVION A ON V.NUMPAV = A.NUMPAV;
Query OK, 0 rows affected (0.006 sec)

MariaDB [sql_projet]> V
-> SELECT * FROM PILOTE WHERE SALAIRE > 15000;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 'V
SELECT * FROM PILOTE WHERE SALAIRE > 15000' at line 1
MariaDB [sql_projet]> CREATE VIEW V_PIL_RICHES AS
-> SELECT * FROM PILOTE WHERE SALAIRE > 15000;
Query OK, 0 rows affected (0.004 sec)

MariaDB [sql_projet]> SELECT * FROM V_PIL_RICHES;
+-----+-----+-----+-----+
| NUMPIL | NOMPIL | VILLE | SALAIRE |
+-----+-----+-----+-----+
| 1 | Alami Amine | Casablanca | 25000.00 |
| 2 | Bennani Youssef | Meknes | 22000.00 |
| 3 | Chraïbi Omar | Marrakech | 18000.00 |
| 4 | Daoudi Mehdi | Rabat | 21000.00 |
| 5 | El Fassi Karim | Fes | 19500.00 |
| 6 | Filali Hamza | Meknes | 23000.00 |
| 7 | Ghazali Anas | Agadir | 17000.00 |
| 8 | Habibi Saïd | Tanger | 20500.00 |
| 9 | Idrissi Khalid | Oujda | 19000.00 |
| 10 | Jebli Hassan | Guelmim | 16000.00 |
| 11 | Kabbaj Ali | Casablanca | 27000.00 |
| 12 | Lahlou Driss | Meknes | 24000.00 |
| 13 | Mansouri Walid | Marrakech | 21500.00 |
| 14 | Naji Brahim | Rabat | 18500.00 |
| 15 | Ouazzani Adil | Fes | 20000.00 |
| 16 | Pacha Reda | Agadir | 17500.00 |
| 17 | Qasmi Mourad | Tanger | 22500.00 |
| 18 | Rami Zakaria | Oujda | 19800.00 |
| 19 | Slaoui Fouad | Guelmim | 16500.00 |
| 20 | Tazi Ismail | Casablanca | 26000.00 |
| 21 | Ummadi Yassine | Marrakech | 20200.00 |
| 22 | Vadi Younes | Meknes | 18800.00 |
| 23 | Wahbi Othman | Rabat | 23500.00 |
| 24 | Xenifi Samy | Fes | 19200.00 |
| 25 | Yacoubi Nabil | Agadir | 17800.00 |
| 26 | Ziani Jalal | Tanger | 21200.00 |
| 27 | Ait Ali Kamal | Oujda | 18200.00 |
| 28 | Berrada Omar | Guelmim | 15800.00 |
| 29 | Chahid Amine | Casablanca | 24500.00 |
| 30 | Didi Hassan | Marrakech | 19700.00 |
+-----+-----+-----+-----+
30 rows in set (0.003 sec)

MariaDB [sql_projet]>
```

Question 25

Supprimer une vue.

SQL:

```
DROP VIEW vue_pilotes_riches;
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
Query OK, 0 rows affected (0.004 sec)

MariaDB [sql_projet]> SELECT * FROM V_PIL_RICHES;
+-----+-----+-----+-----+
| NOMPIL | NOMPIL | VILLE | SALAIRE |
+-----+-----+-----+-----+
| 1 | Alami Amine | Casablanca | 25000.00 |
| 2 | Bennani Youssef | Meknes | 22000.00 |
| 3 | Chraïbi Omar | Marrakech | 18000.00 |
| 4 | Daoudi Mehdi | Rabat | 21000.00 |
| 5 | El Fassi Karim | Fes | 19500.00 |
| 6 | Filali Hamza | Meknes | 23000.00 |
| 7 | Ghazali Anas | Agadir | 17000.00 |
| 8 | Habibi Said | Tanger | 20500.00 |
| 9 | Idrissi Khalid | Oujda | 19000.00 |
| 10 | Jebli Hassan | Guelmim | 16000.00 |
| 11 | Kabbaj Ali | Casablanca | 27000.00 |
| 12 | Lahlou Driss | Meknes | 24000.00 |
| 13 | Mansouri Walid | Marrakech | 21500.00 |
| 14 | Naji Brahim | Rabat | 18500.00 |
| 15 | Ouazzani Adil | Fes | 20000.00 |
| 16 | Pacha Reda | Agadir | 17500.00 |
| 17 | Qasmi Mourad | Tanger | 22500.00 |
| 18 | Rami Zakaria | Oujda | 19800.00 |
| 19 | Slaoui Fouad | Guelmim | 16500.00 |
| 20 | Tazi Ismail | Casablanca | 26000.00 |
| 21 | Ummadi Yassine | Marrakech | 20200.00 |
| 22 | Vadi Younes | Meknes | 18800.00 |
| 23 | Wahbi Othman | Rabat | 23500.00 |
| 24 | Xenifi Samy | Fes | 10200.00 |
| 25 | Yacoubi Nabil | Agadir | 17800.00 |
| 26 | Ziani Jalal | Tanger | 21200.00 |
| 27 | Ait Ali Kamal | Oujda | 18200.00 |
| 28 | Berrada Omar | Guelmim | 15800.00 |
| 29 | Chahid Amine | Casablanca | 24500.00 |
| 30 | Didi Hassan | Marrakech | 19700.00 |
+-----+-----+-----+-----+
30 rows in set (0.003 sec)

MariaDB [sql_projet]> DROP VIEW V_PIL_RICHES;
Query OK, 0 rows affected (0.011 sec)

MariaDB [sql_projet]> _
```

Question 26

Créer une vue contenant uniquement les vols au départ de Marrakech.

SQL:

```
CREATE VIEW vue_vols_marrakech AS SELECT * FROM VOL WHERE ville_dep = 'Marrakech';
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
Query OK, 0 rows affected (0.004 sec)

MariaDB [sql_projet]> SELECT * FROM V_PIL_RICHES;
+-----+-----+-----+-----+
| NUMPIL | NOMPIL | VILLE | SALAIRE |
+-----+-----+-----+-----+
| 1 | Alami Amine | Casablanca | 25000.00 |
| 2 | Bennani Youssef | Meknes | 22000.00 |
| 3 | Chraïbi Omar | Marrakech | 18000.00 |
| 4 | Daoudi Mehdi | Rabat | 21000.00 |
| 5 | El Fassi Karim | Fes | 19500.00 |
| 6 | Filali Hamza | Meknes | 23000.00 |
| 7 | Ghazali Anas | Agadir | 17000.00 |
| 8 | Habibi Saïd | Tanger | 20500.00 |
| 9 | Idrissi Khalid | Oujda | 19000.00 |
| 10 | Jebli Hassan | Guelmim | 16000.00 |
| 11 | Kabbaï Ali | Casablanca | 27000.00 |
| 12 | Lahlou Driss | Meknes | 24000.00 |
| 13 | Mansouri Walid | Marrakech | 21500.00 |
| 14 | Naïji Brahim | Rabat | 18500.00 |
| 15 | Ouazzani Adil | Fes | 20000.00 |
| 16 | Pacha Reda | Agadir | 17500.00 |
| 17 | Qasmi Mourad | Tanger | 22500.00 |
| 18 | Rami Zakaria | Oujda | 19800.00 |
| 19 | Slaoui Fouad | Guelmim | 16500.00 |
| 20 | Tazi Ismail | Casablanca | 26000.00 |
| 21 | Ummadi Yassine | Marrakech | 20200.00 |
| 22 | Vadi Younes | Meknes | 18800.00 |
| 23 | Wahbi Othman | Rabat | 23500.00 |
| 24 | Xenifi Samy | Fes | 19200.00 |
| 25 | Yecoubi Nabil | Agadir | 17800.00 |
| 26 | Ziani Jalal | Tanger | 21200.00 |
| 27 | Ait Ali Kamal | Oujda | 18200.00 |
| 28 | Berrada Omar | Guelmim | 15800.00 |
| 29 | Chahid Amine | Casablanca | 24500.00 |
| 30 | Didi Hassan | Marrakech | 19700.00 |
+-----+-----+-----+-----+
30 rows in set (0.003 sec)

MariaDB [sql_projet]> DROP VIEW V_PIL_RICHES;
Query OK, 0 rows affected (0.011 sec)

MariaDB [sql_projet]> CREATE VIEW V_VOL_MARRAKECH AS
-> SELECT * FROM VOL WHERE VILLE_DEP = 'Marrakech';
Query OK, 0 rows affected (0.004 sec)

MariaDB [sql_projet]>
```

Question 27

Créer un utilisateur nommé user_air.

SQL:

```
CREATE USER user_air;
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
Query OK, 0 rows affected (0.004 sec)

MariaDB [sql_projet]> SELECT * FROM V_PIL_RICHES;
+-----+-----+-----+-----+
| NOMPIL | NOMPIL | VILLE | SALAIRE |
+-----+-----+-----+-----+
| 1 | Alami Amine | Casablanca | 25000.00 |
| 2 | Bennani Youssef | Meknes | 22000.00 |
| 3 | Chraïbi Omar | Marrakech | 18000.00 |
| 4 | Daoudi Mehdi | Rabat | 21000.00 |
| 5 | El Fassi Karim | Fes | 19500.00 |
| 6 | Filali Hamza | Meknes | 23000.00 |
| 7 | Ghazali Anas | Agadir | 17000.00 |
| 8 | Habibi Said | Tanger | 20500.00 |
| 9 | Idrissi Khalid | Oujda | 19000.00 |
| 10 | Jebli Hassan | Guelmim | 16000.00 |
| 11 | Kabbaj Ali | Casablanca | 27000.00 |
| 12 | Lahlou Driss | Meknes | 24000.00 |
| 13 | Mansouri Walid | Marrakech | 21500.00 |
| 14 | Naji Brahim | Rabat | 18500.00 |
| 15 | Ouazzani Adil | Fes | 20000.00 |
| 16 | Pacha Reda | Agadir | 17500.00 |
| 17 | Qasmi Mourad | Tanger | 22500.00 |
| 18 | Rami Zakaria | Oujda | 19800.00 |
| 19 | Slaoui Fouad | Guelmim | 16500.00 |
| 20 | Tazi Ismail | Casablanca | 26000.00 |
| 21 | Ummadi Yassine | Marrakech | 20200.00 |
| 22 | Vadi Younes | Meknes | 18800.00 |
| 23 | Wahbi Othman | Rabat | 23500.00 |
| 24 | Xenifi Samy | Fes | 19200.00 |
| 25 | Yacoubi Nabil | Agadir | 17800.00 |
| 26 | Ziani Jalal | Tanger | 21200.00 |
| 27 | Ait Ali Kamal | Oujda | 18200.00 |
| 28 | Berrada Omar | Guelmim | 15800.00 |
| 29 | Chahid Amine | Casablanca | 24500.00 |
| 30 | Didi Hassan | Marrakech | 19700.00 |
+-----+-----+-----+-----+
30 rows in set (0.003 sec)

MariaDB [sql_projet]> DROP VIEW V_PIL_RICHES;
Query OK, 0 rows affected (0.011 sec)

MariaDB [sql_projet]> CREATE VIEW V_VOL_MARRAKECH AS
-> SELECT * FROM VOL WHERE VILLE_DEP = 'Marrakech';
Query OK, 0 rows affected (0.004 sec)

MariaDB [sql_projet]> CREATE USER 'user_air'@'localhost' IDENTIFIED BY 'pass123';
Query OK, 0 rows affected (0.004 sec)

MariaDB [sql_projet]> _
```

Question 28

Donner à cet utilisateur le droit de consulter la table PILOTE.

SQL:

```
GRANT SELECT ON PILOTE TO user_air;
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
MariaDB [sql_projet]> SELECT * FROM V_PIL_RICHES;
+-----+-----+-----+-----+
| NOMPIL | NOMPIL | VILLE | SALAIRE |
+-----+-----+-----+-----+
| 1 | Alami Amine | Casablanca | 25000.00 |
| 2 | Bennani Youssef | Meknes | 22000.00 |
| 3 | Chraïbi Omar | Marrakech | 18000.00 |
| 4 | Daoudi Mehdi | Rabat | 21000.00 |
| 5 | El Fassi Karim | Fes | 19500.00 |
| 6 | Filali Hamza | Meknes | 23000.00 |
| 7 | Ghazali Anas | Agadir | 17000.00 |
| 8 | Habibi Saïd | Tanger | 20500.00 |
| 9 | Idrissi Khalid | Oujda | 19000.00 |
| 10 | Jebli Hassan | Guelmim | 16000.00 |
| 11 | Kabbaj Ali | Casablanca | 27000.00 |
| 12 | Lahlou Driss | Meknes | 24000.00 |
| 13 | Mansouri Walid | Marrakech | 21500.00 |
| 14 | Naji Brahim | Rabat | 18500.00 |
| 15 | Ouazzani Adil | Fes | 20000.00 |
| 16 | Pacha Reda | Agadir | 17500.00 |
| 17 | Qasmi Mourad | Tanger | 22500.00 |
| 18 | Rami Zakaria | Oujda | 19800.00 |
| 19 | Slaoui Fouad | Guelmim | 16500.00 |
| 20 | Tazi Ismail | Casablanca | 26000.00 |
| 21 | Ummadi Yassine | Marrakech | 20200.00 |
| 22 | Vadi Younes | Meknes | 18800.00 |
| 23 | Wahbi Othman | Rabat | 23500.00 |
| 24 | Xenifi Samy | Fes | 19200.00 |
| 25 | Yacoubi Nabil | Agadir | 17800.00 |
| 26 | Ziani Jalal | Tanger | 21200.00 |
| 27 | Ait Ali Kamal | Oujda | 18200.00 |
| 28 | Berrada Omar | Guelmim | 15800.00 |
| 29 | Chahid Amine | Casablanca | 24500.00 |
| 30 | Didi Hassan | Marrakech | 19700.00 |
+-----+-----+-----+-----+
30 rows in set (0.003 sec)

MariaDB [sql_projet]> DROP VIEW V_PIL_RICHES;
Query OK, 0 rows affected (0.011 sec)

MariaDB [sql_projet]> CREATE VIEW V_VOL_MARRAKECH AS
-> SELECT * FROM VOL WHERE VILLE_DEP = 'Marrakech';
Query OK, 0 rows affected (0.004 sec)

MariaDB [sql_projet]> CREATE USER 'user_air'@'localhost' IDENTIFIED BY 'pass123';
Query OK, 0 rows affected (0.004 sec)

MariaDB [sql_projet]> GRANT SELECT ON sql_projet.PILOTE TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.002 sec)

MariaDB [sql_projet]> _
```

Question 29

Donner les droits d'insertion sur la table VOL.

SQL:

```
GRANT INSERT ON VOL TO user_air;
```

Résultat:

```
C:\Windows\system32\cmd.exe - mysql -u root -p
+-----+-----+-----+-----+
| NUMPIL | NOMPIL | VILLE | SALAIRE |
+-----+-----+-----+-----+
| 1 | Alami Amine | Casablanca | 25000.00 |
| 2 | Bennani Youssef | Meknes | 22000.00 |
| 3 | Chraïbi Omar | Marrakech | 18000.00 |
| 4 | Daoudi Mehdi | Rabat | 21000.00 |
| 5 | El Fassi Karim | Fes | 19500.00 |
| 6 | Filali Hamza | Meknes | 23000.00 |
| 7 | Ghazali Anas | Agadir | 17000.00 |
| 8 | Habibi Said | Tanger | 20500.00 |
| 9 | Idrissi Khalid | Oujda | 19000.00 |
| 10 | Jebli Hassan | Guelmim | 16000.00 |
| 11 | Kabbaj Ali | Casablanca | 27000.00 |
| 12 | Lahlou Driss | Meknes | 24000.00 |
| 13 | Mansour Walid | Marrakech | 21500.00 |
| 14 | Maji Brahim | Rabat | 18500.00 |
| 15 | Ouazzani Adil | Fes | 20000.00 |
| 16 | Pacha Reda | Agadir | 17500.00 |
| 17 | Qasmi Mourad | Tanger | 22500.00 |
| 18 | Rami Zakaria | Oujda | 19800.00 |
| 19 | Slaoui Fouad | Guelmim | 16500.00 |
| 20 | Tazi Ismail | Casablanca | 26000.00 |
| 21 | Ummadi Yassine | Marrakech | 20200.00 |
| 22 | Vadi Younes | Meknes | 18800.00 |
| 23 | Wahbi Othman | Rabat | 23500.00 |
| 24 | Xenifi Samy | Fes | 19200.00 |
| 25 | Yacoubi Nabil | Agadir | 17800.00 |
| 26 | Ziani Jalal | Tanger | 21200.00 |
| 27 | Ait Ali Kamal | Oujda | 18200.00 |
| 28 | Berrada Omar | Guelmim | 15800.00 |
| 29 | Chahid Amine | Casablanca | 24500.00 |
| 30 | Didi Hassan | Marrakech | 19700.00 |
+-----+-----+-----+-----+
30 rows in set (0.003 sec)

MariaDB [sql_projet]> DROP VIEW V_PIL_RICHES;
Query OK, 0 rows affected (0.011 sec)

MariaDB [sql_projet]> CREATE VIEW V_VOL_MARRAKECH AS
-> SELECT * FROM VOL WHERE VILLE_DEP = 'Marrakech';
Query OK, 0 rows affected (0.004 sec)

MariaDB [sql_projet]> CREATE USER 'user_air'@'localhost' IDENTIFIED BY 'pass123';
Query OK, 0 rows affected (0.004 sec)

MariaDB [sql_projet]> GRANT SELECT ON sql_projet.PILOTE TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.002 sec)

MariaDB [sql_projet]> GRANT INSERT ON sql_projet.VOL TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.002 sec)

MariaDB [sql_projet]> _
```

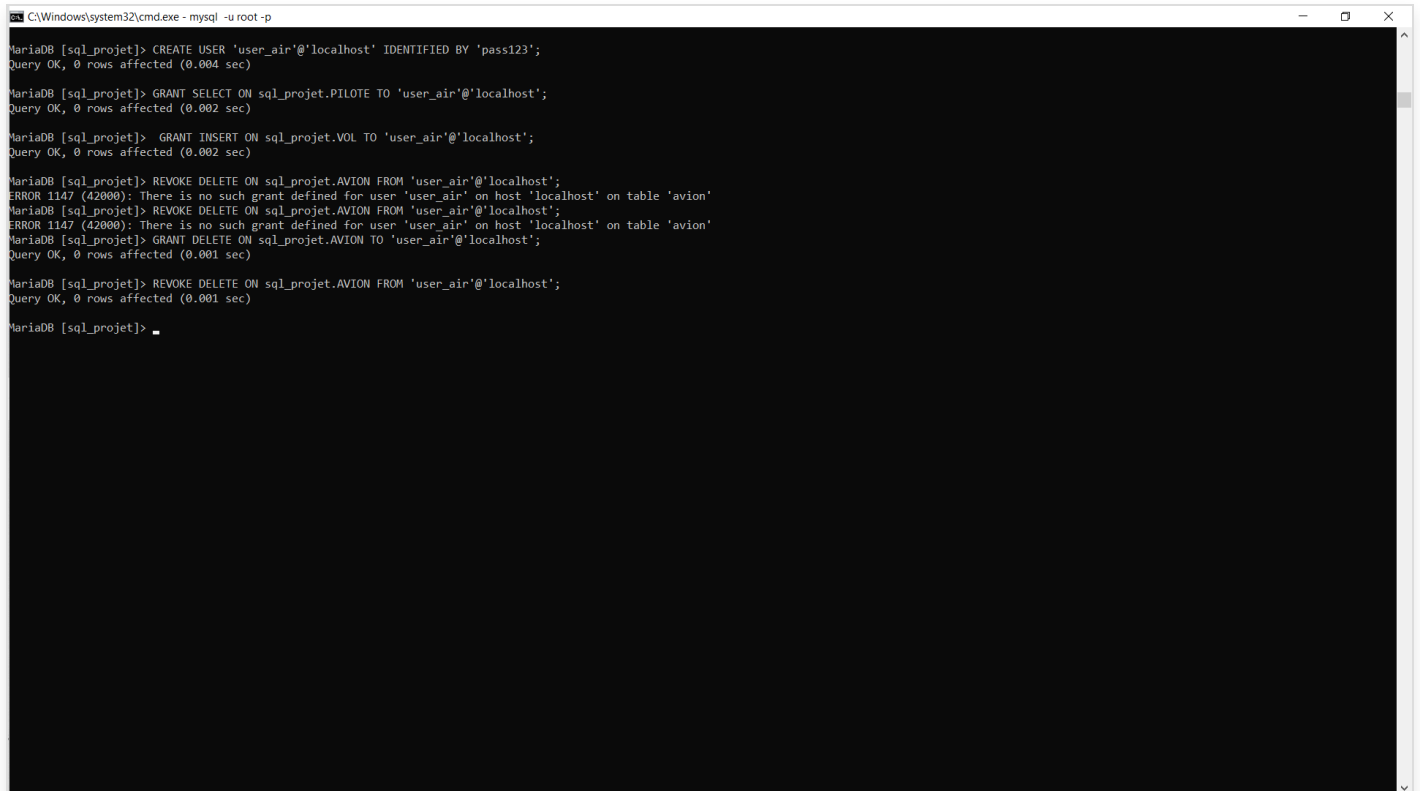

Question 30

Retirer le droit de suppression sur la table AVION.

SQL:

```
REVOKE DELETE ON AVION FROM user_air;
```

Résultat:



```
C:\Windows\system32\cmd.exe - mysql -u root -p

MariaDB [sql_projet]> CREATE USER 'user_air'@'localhost' IDENTIFIED BY 'pass123';
Query OK, 0 rows affected (0.004 sec)

MariaDB [sql_projet]> GRANT SELECT ON sql_projet.PILOTE TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.002 sec)

MariaDB [sql_projet]> GRANT INSERT ON sql_projet.VOL TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.002 sec)

MariaDB [sql_projet]> REVOKE DELETE ON sql_projet.AVION FROM 'user_air'@'localhost';
ERROR 1147 (42000): There is no such grant defined for user 'user_air' on host 'localhost' on table 'avion'
MariaDB [sql_projet]> REVOKE DELETE ON sql_projet.AVION FROM 'user_air'@'localhost';
ERROR 1147 (42000): There is no such grant defined for user 'user_air' on host 'localhost' on table 'avion'
MariaDB [sql_projet]> GRANT DELETE ON sql_projet.AVION TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [sql_projet]> REVOKE DELETE ON sql_projet.AVION FROM 'user_air'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [sql_projet]> .
```

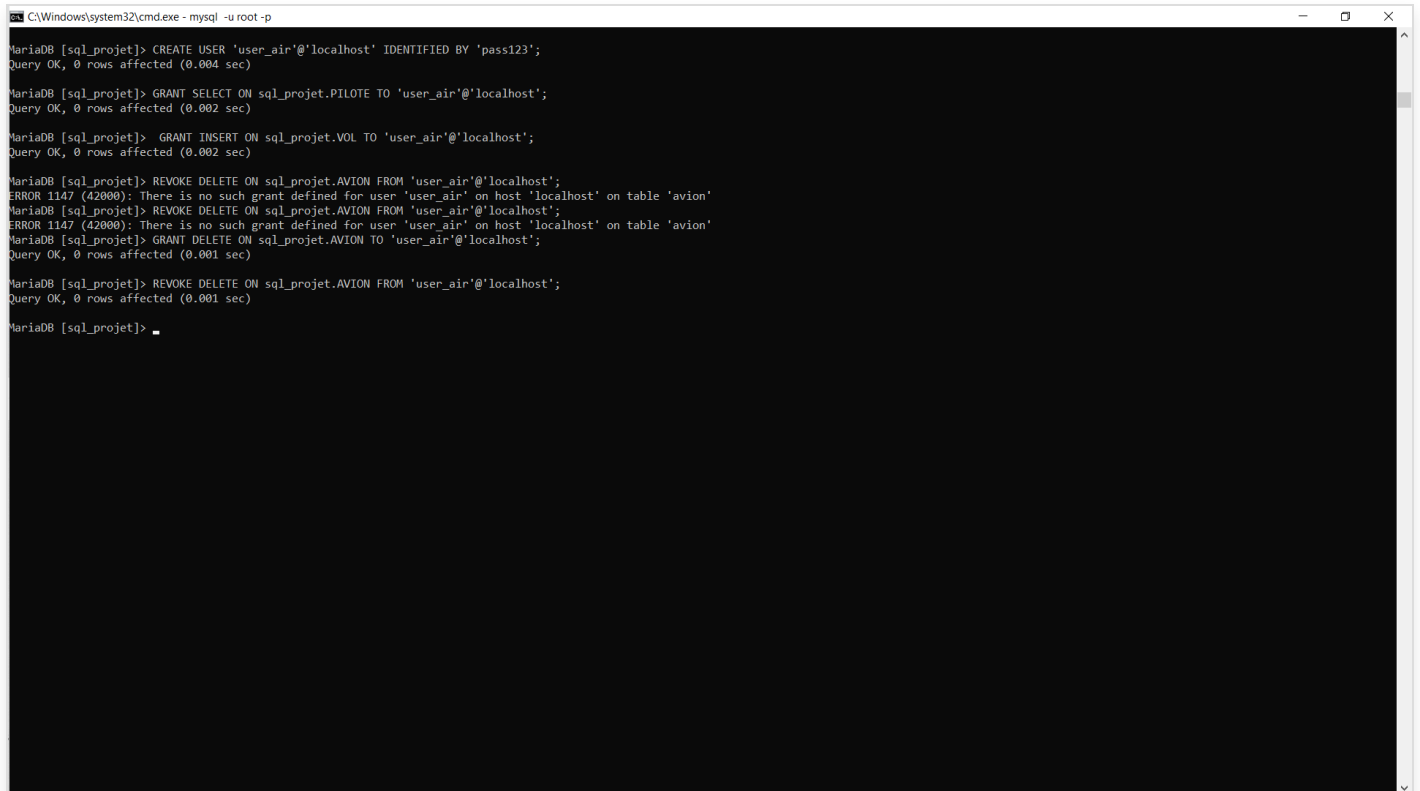
Question 31

Donner tous les privilèges sur la base BD_AIR_MAROC.

SQL:

```
GRANT ALL PRIVILEGES ON BD_AIR_MAROC.* TO user_air;
```

Résultat:



```
C:\Windows\system32\cmd.exe - mysql -u root -p

MariaDB [sql_projet]> CREATE USER 'user_air'@'localhost' IDENTIFIED BY 'pass123';
Query OK, 0 rows affected (0.004 sec)

MariaDB [sql_projet]> GRANT SELECT ON sql_projet.PILOTE TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.002 sec)

MariaDB [sql_projet]> GRANT INSERT ON sql_projet.VOL TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.002 sec)

MariaDB [sql_projet]> REVOKE DELETE ON sql_projet.AVION FROM 'user_air'@'localhost';
ERROR 1147 (42000): There is no such grant defined for user 'user_air' on host 'localhost' on table 'avion'
MariaDB [sql_projet]> REVOKE DELETE ON sql_projet.AVION FROM 'user_air'@'localhost';
ERROR 1147 (42000): There is no such grant defined for user 'user_air' on host 'localhost' on table 'avion'
MariaDB [sql_projet]> GRANT DELETE ON sql_projet.AVION TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [sql_projet]> REVOKE DELETE ON sql_projet.AVION FROM 'user_air'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [sql_projet]> .
```

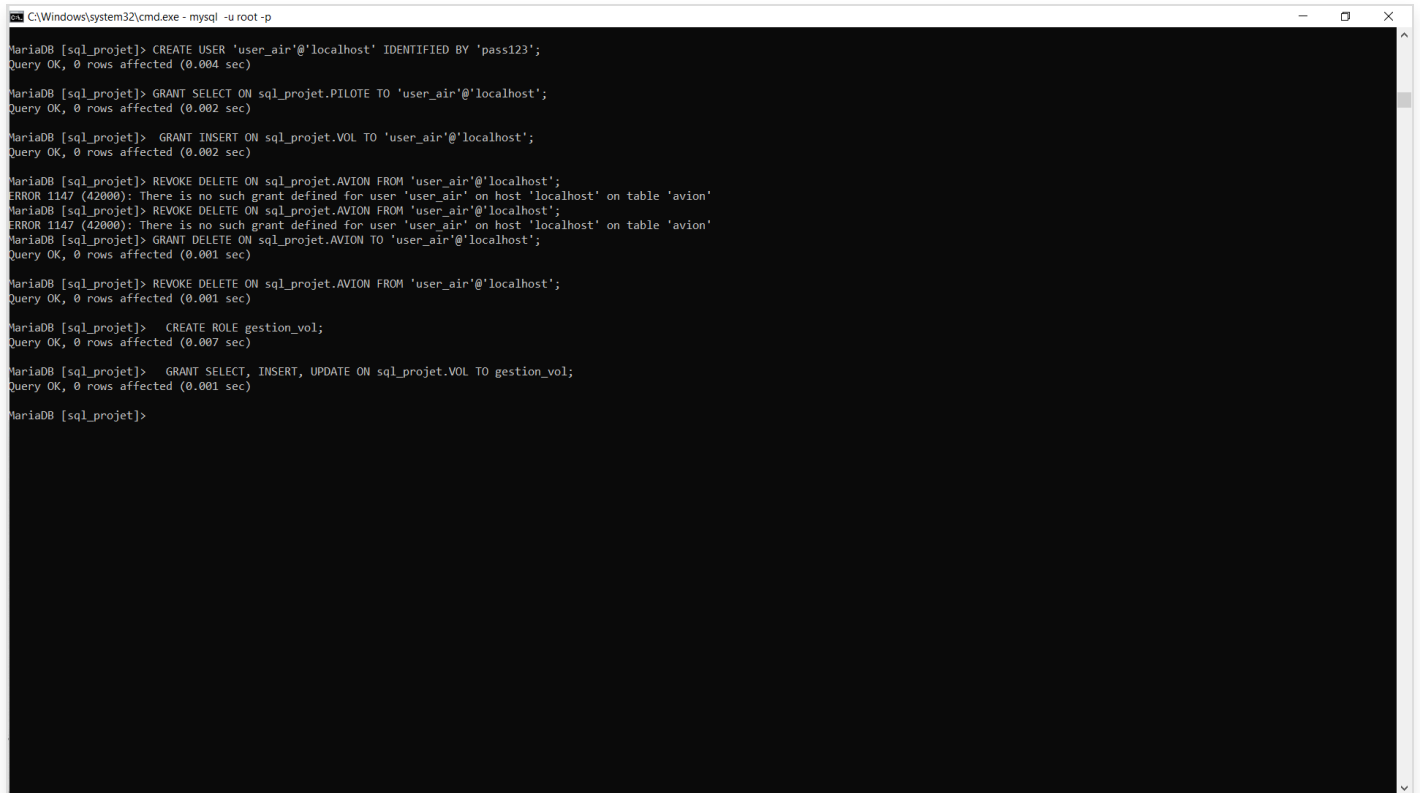
Question 32

Créer un rôle gestion_vol et lui attribuer des droits sur la table VOL.

SQL:

```
CREATE ROLE gestion_vol;\nGRANT ALL PRIVILEGES ON VOL TO gestion_vol;
```

Résultat:



```
C:\Windows\system32\cmd.exe - mysql -u root -p

MariaDB [sql_projet]> CREATE USER 'user_air'@'localhost' IDENTIFIED BY 'pass123';
Query OK, 0 rows affected (0.004 sec)

MariaDB [sql_projet]> GRANT SELECT ON sql_projet.PILOTE TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.002 sec)

MariaDB [sql_projet]> GRANT INSERT ON sql_projet.VOL TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.002 sec)

MariaDB [sql_projet]> REVOKE DELETE ON sql_projet.AVION FROM 'user_air'@'localhost';
ERROR 1147 (42000): There is no such grant defined for user 'user_air' on host 'localhost' on table 'avion'
MariaDB [sql_projet]> REVOKE DELETE ON sql_projet.AVION FROM 'user_air'@'localhost';
ERROR 1147 (42000): There is no such grant defined for user 'user_air' on host 'localhost' on table 'avion'
MariaDB [sql_projet]> GRANT DELETE ON sql_projet.AVION TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [sql_projet]> REVOKE DELETE ON sql_projet.AVION FROM 'user_air'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [sql_projet]> CREATE ROLE gestion_vol;
Query OK, 0 rows affected (0.007 sec)

MariaDB [sql_projet]> GRANT SELECT, INSERT, UPDATE ON sql_projet.VOL TO gestion_vol;
Query OK, 0 rows affected (0.001 sec)

MariaDB [sql_projet]>
```

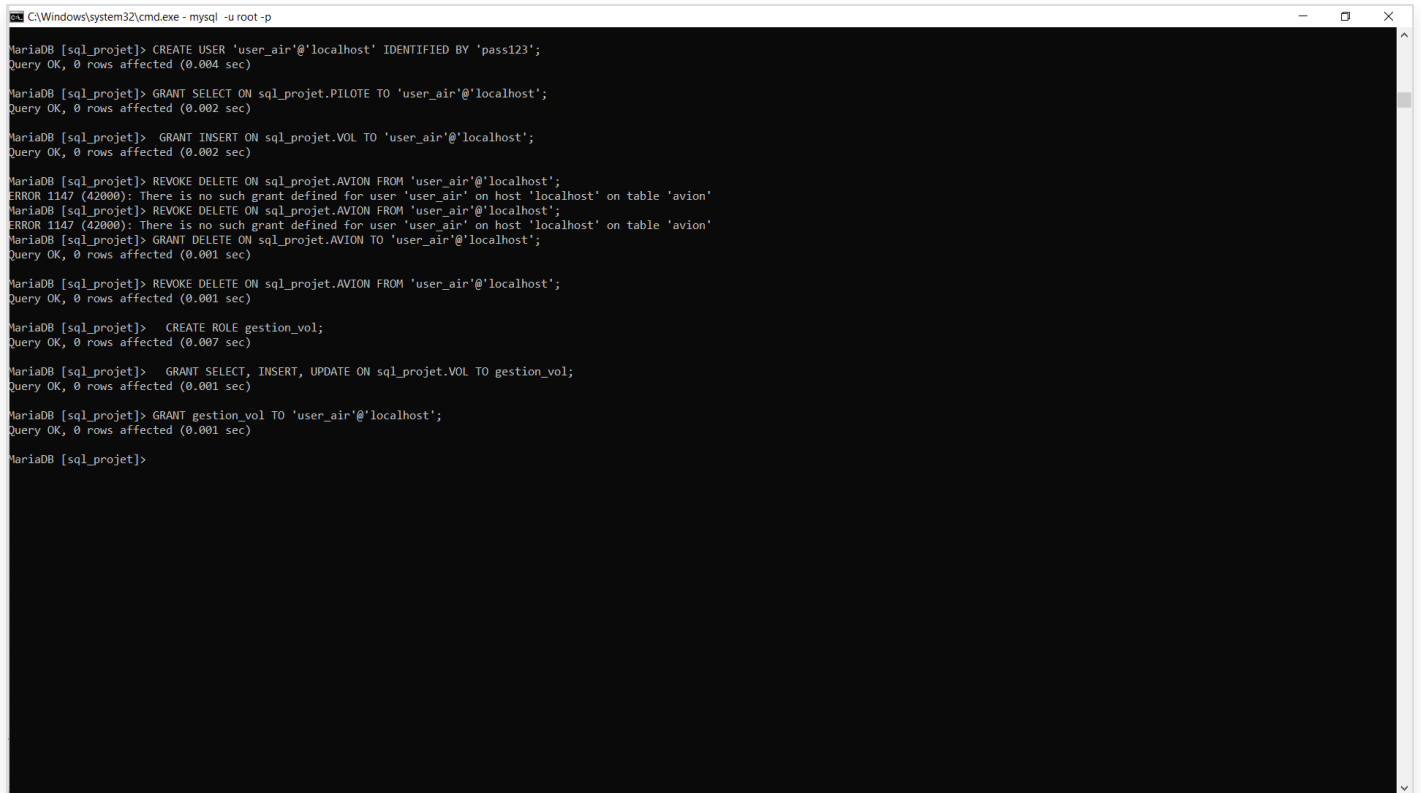
Question 33

Affecter ce rôle à un utilisateur.

SQL:

```
GRANT gestion_vol TO user_air;
```

Résultat:



```
C:\Windows\system32\cmd.exe - mysql -u root -p
MariaDB [sql_projet]> CREATE USER 'user_air'@'localhost' IDENTIFIED BY 'pass123';
Query OK, 0 rows affected (0.004 sec)

MariaDB [sql_projet]> GRANT SELECT ON sql_projet.PILOTE TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.002 sec)

MariaDB [sql_projet]> GRANT INSERT ON sql_projet.VOL TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.002 sec)

MariaDB [sql_projet]> REVOKE DELETE ON sql_projet.AVION FROM 'user_air'@'localhost';
ERROR 1147 (42000): There is no such grant defined for user 'user_air' on host 'localhost' on table 'avion'
MariaDB [sql_projet]> REVOKE DELETE ON sql_projet.AVION FROM 'user_air'@'localhost';
ERROR 1147 (42000): There is no such grant defined for user 'user_air' on host 'localhost' on table 'avion'
MariaDB [sql_projet]> GRANT DELETE ON sql_projet.AVION TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [sql_projet]> REVOKE DELETE ON sql_projet.AVION FROM 'user_air'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [sql_projet]> CREATE ROLE gestion_vol;
Query OK, 0 rows affected (0.007 sec)

MariaDB [sql_projet]> GRANT SELECT, INSERT, UPDATE ON sql_projet.VOL TO gestion_vol;
Query OK, 0 rows affected (0.001 sec)

MariaDB [sql_projet]> GRANT gestion_vol TO 'user_air'@'localhost';
Query OK, 0 rows affected (0.001 sec)

MariaDB [sql_projet]>
```